

Dengue raw data cleaner

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Load libraries

```
.pkgs <- c(
  "tidyverse",
  "janitor",
  "fs",
  "readxl",
  "skimr",
  "stringi",
  "patchwork"
)
xfun::pkg_attach(.pkgs)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.6
v forcats    1.0.1      v stringr    1.6.0
v ggplot2    4.0.1      v tibble     3.3.0
v lubridate  1.9.4      v tidyr      1.3.2
v purrr      1.2.0

-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
conflicted

Attaching package: 'janitor'
```

The following objects are masked from 'package:stats':

```
chisq.test, fisher.test
```

```
theme_set(theme_bw())
```

Data ingestion

Read raw excel files and quickly clean column names with `janitor::clean_names()`

```
xlsx_raws <- list.files(
  "incidence_data_geocoded",
  pattern = "xlsx",
  full.names = TRUE
) %>%
  map(
    ~ map(excel_sheets(.x), \(sh) {
      suppressWarnings(read_excel(.x, sheet = sh)) %>% clean_names()
    })
  ) %>%
  set_names(NULL)
```

Our columns of interest are: “sex”, “age”, “date of admission”, “district (of patient’s residential address)”, “commune (of patient’s residential address)”, “hospital”, “icd”, “in-/out-patient”

EDA and data cleaning

There are 3 excel files

- First file contains cases from 2000-2016, a separate sheet for each year.
- Second file contains cases from 2017-2025, a separate sheet for each year
- Third file contains cases from August 2025 (after major change in admin level).

Data from 2000-2016

We can look at mismatched columns for all the sheets

```
compare_df_cols(xlsx_rows[[1]], return = "mismatch")
```

```

column_name  xlsx_rows[[1]]_1 xlsx_rows[[1]]_2 xlsx_rows[[1]]_3
1  ng_tuvong          logical  POSIXct, POSIXt          logical
  xlsx_rows[[1]]_4 xlsx_rows[[1]]_5 xlsx_rows[[1]]_6 xlsx_rows[[1]]_7
1          <NA>          <NA>  POSIXct, POSIXt          <NA>
  xlsx_rows[[1]]_8 xlsx_rows[[1]]_9 xlsx_rows[[1]]_10 xlsx_rows[[1]]_11
1  POSIXct, POSIXt  POSIXct, POSIXt          logical          <NA>
  xlsx_rows[[1]]_12 xlsx_rows[[1]]_13 xlsx_rows[[1]]_14 xlsx_rows[[1]]_15
1  POSIXct, POSIXt          <NA>          <NA>          <NA>
  xlsx_rows[[1]]_16 xlsx_rows[[1]]_17
1          <NA>          <NA>

```

Mismatched columns `cd_ravien`, `ng_bc`, `ng_tuvong` are insignificant, so we will go ahead with row binding all the sheets together

`xlsx_rows[-length(xlsx_rows)]` means look at all sheets from first file, ignore last sheet (i.e. second file)

```
raw_2000_2016 <- xlsx_rows[[1]] %>% bind_rows()
```

Skimming to see what's going on in the data

```
skim(raw_2000_2016)
```

Table 1: Data summary

Name	raw_2000_2016
Number of rows	147927
Number of columns	24
Column type frequency:	
character	14
numeric	5
POSIXct	5
Group variables	None

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
maso	43969	0.70	1	15	0	88299	0
hoten	0	1.00	3	39	0	110896	0
gioi	6351	0.96	1	1	0	7	0
diachi	10210	0.93	2	92	0	129420	0
px	0	1.00	2	16	0	171	0
qh	0	1.00	2	10	0	24	0
cd_ravien	96007	0.35	14	14	0	1	0
nguong_du_lieu	0	1.00	7	18	0	40	0
do_sxh	31125	0.79	1	1	0	15	0
m_icd	7495	0.95	3	6	0	37	0
id	43969	0.70	1	15	0	88299	0
diachi_api	14211	0.90	29	210	0	94353	0
px_moi	14211	0.90	6	21	0	103	0
nasos	144843	0.02	4	10	0	3082	0

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
tuoi	22100	0.85	16.33	240.80	-	8.00	13.00	22.00	2015.00	34669.00
ng_sinh	32149	0.78	1994.34	251.00	0.00	1988.00	1996.00	2003.00	36685.00	
ma_tinh	0	1.00	701.00	0.00	701.00	701.00	701.00	701.00	701.00	
long	14211	0.90	106.67	0.06	106.37	106.64	106.67	106.70	106.98	
lat	14211	0.90	10.78	0.06	10.38	10.75	10.78	10.81	11.15	

Variable type: POSIXct

skim_variable	n_missing	complete_rate	min	max	median	n_unique
ng_vaovien	0	1.00	2000-01-01	2016-12-31 22:14:28	2010-04-19	8103
ng_ravien	1411	0.99	1900-01-22	2017-03-16 00:00:00	2010-07-21	7109
ng_tuvong	147888	0.00	2001-03-20	2011-11-25 00:00:00	2008-08-24	39
ng_bc	56911	0.62	2000-01-07	2017-03-22 00:00:00	2007-11-29	4101

skim_vari- able	n_miss- ing	com- plete_rate	min	max	median	n_unique
ng_nhap	364	1.00	2000-01- 07	2017-03-24 00:00:00	2010-05- 07	2033

The actual columns based on our columns of interest are (with `complete_rate`):

- `sex = gioi` (0.957)
- `age = tuoi` (0.851); `ng_sinh` (0.783); year of birth might be in `hoten` col
- `date of admission = ng_vaovien` (1)
- `district = qh` (1)
- `commune = px` (1)
- `new_commune = px_moi` (0.904)
- `hospital = nguon_du_lieu` (1)
- `icd = m_icd` (0.949)
- `in-patient` = this is all in-patient data

Selecting columns of interest

```
s1_2000_2016 <- raw_2000_2016 %>%
  select(
    hoten,
    gioi,
    tuoi,
    ng_sinh,
    ng_vaovien,
    qh,
    px,
    px_moi,
    nguon_du_lieu,
    m_icd
  )
# s1_2000_2016
```

Keeping track of raw number of rows at start to see how much lost during data cleaning

```
start_nrow <- nrow(s1_2000_2016)
start_nrow
```

```
[1] 147927
```

Fix age if possible

Some year of births (YOBs) are stored in the name (**hoten**) column, let's see how much of the data is like this

```
s1_2000_2016 %>%
  select(hoten, tuoi, ng_sinh) %>%
  filter(is.na(tuoi)) %>%
  rowwise() %>%
  mutate(
    no_age = any(
      varhandle::check.numeric(hoten),
      !is.na(ng_sinh)
    )
  ) %>%
  tabyl(no_age)
```

no_age	n	percent
FALSE	21898	0.990859729
TRUE	202	0.009140271

Less than 1% of the data has YOB in the name column and able to get age from YOB, so it's not worth going deeper.

For now, people with no age will have NA as their age

Rename columns

```
s2_2000_2016 <- s1_2000_2016 %>%
  select(-hoten, -ng_sinh) %>%
  rename(
    sex = gioi,
    age = tuoi,
    date = ng_vaovien,
```

```

    district = qh,
    commune = px,
    new_commune = px_moi,
    hospital = nguon_du_lieu,
    icd = m_icd
  ) %>%
  mutate(
    in_out_patient = "in-patient"
  )
# s2_2000_2016

```

Fix dates data

Date data is in `datetime`, convert to `date` only

```

s3_2000_2016 <- s2_2000_2016 %>%
  mutate(date = convert_to_date(date))
# s3_2000_2016

```

```
skim(s3_2000_2016)
```

Table 2: Data summary

Name	s3_2000_2016
Number of rows	147927
Number of columns	9
Column type frequency:	
character	7
Date	1
numeric	1
Group variables	None

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
sex	6351	0.96	1	1	0	7	0
district	0	1.00	2	10	0	24	0

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
commune	0	1.00	2	16	0	171	0
new_commune	14211	0.90	6	21	0	103	0
hospital	0	1.00	7	18	0	40	0
icd	7495	0.95	3	6	0	37	0
in_out_patient	0	1.00	10	10	0	1	0

Variable type: Date

skim_variable	n_missing	complete_rate	min	max	median	n_unique
date	0	1	2000-01-01	2016-12-31	2010-04-19	6184

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
age	22100	0.85	16.33	240.8	-	8	13	22	2015	34669

We see that there are 7 unique sexes, 37 unique ICDs, negative age. Let's clean all of this

Fix sexes

Look into sex column first

```
s3_2000_2016 %>% tabyl(sex)
```

```
sex      n      percent valid_percent
1         1 6.760091e-06 7.063344e-06
G 59814 4.043481e-01 4.224869e-01
I  9073 6.133431e-02 6.408572e-02
N   890 6.016481e-03 6.286376e-03
T 71786 4.852799e-01 5.070492e-01
g         4 2.704036e-05 2.825338e-05
t         8 5.408073e-05 5.650675e-05
<NA> 6351 4.293334e-02          NA
```

Impossible to know what “I” and “N”, treat these as NA instead

Relabel `sex` column

```
s3_2000_2016 <- s3_2000_2016 |>
  mutate(
    sex = tolower(sex),
    sex = case_when(
      sex == "g" ~ "f",
      sex == "t" ~ "m",
      .default = NA
    )
  )
```

Fix ICD

```
s3_2000_2016 %>%
  mutate(year = year(date), icd = tolower(icd)) %>%
  tabyl(icd, year)
```

icd	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013
a90	0	0	0	0	0	0	0	0	0	0	140	937	15	1
a91	375	556	865	2680	3329	2526	2955	1180	3672	1378	38	146	9	20
a91-1	0	30	31	55	64	52	78	129	83	137	225	30	4	0
a91-2	6	456	571	1300	1187	994	1807	2437	2278	2686	2241	480	11	0
a91-3	0	74	99	325	194	115	151	280	355	389	334	79	1	0
a91-4	0	7	10	15	11	4	16	17	13	35	27	9	0	0
a91-e	0	0	0	0	0	0	0	0	0	0	0	0	2	1
a91.	0	0	0	0	0	0	0	0	0	0	9	5	0	0
a91.]	0	0	0	0	0	0	0	1	0	0	0	0	0	0
a91.`	0	0	0	0	0	0	0	1	0	0	0	0	0	0
a91.0	5	8	4	6	0	1	0	4	0	2	0	0	0	0
a91.1	946	979	657	1769	748	814	947	1368	1516	1319	62	223	0	0
a91.2	647	548	428	1791	1538	1469	2221	3424	5631	4723	200	542	0	0
a91.3	147	133	119	407	227	151	198	251	613	411	21	61	0	0
a91.4	10	11	6	29	13	8	8	11	66	31	1	0	0	0
a91.9	0	0	0	0	0	0	0	0	3	0	0	0	0	0
a91.a	0	0	0	0	0	0	0	0	0	0	0	6113	7407	5310
a91.b	0	0	0	0	0	0	0	0	0	0	0	1149	1400	1081
a91.c	0	0	0	0	0	0	0	0	0	0	0	253	244	156
a91.c1	0	0	0	0	0	0	0	0	0	0	0	317	423	285

a91.c2	0	0	0	0	0	0	0	0	0	0	0	17	30	35
a91.c3	0	0	0	0	0	0	0	0	0	0	0	7	1	3
a91.c4	0	0	0	0	0	0	0	0	0	0	0	10	12	16
a99.1	0	0	0	0	0	0	0	0	0	0	3	30	0	0
a99.2	0	0	0	0	0	0	0	0	0	0	12	478	0	0
a99.3	0	0	0	0	0	0	0	0	0	0	2	41	0	0
a99.4	0	0	0	0	0	0	0	0	0	0	0	5	0	0
<NA>	0	0	0	0	0	0	0	0	0	0	6373	888	233	1
2014	2015	2016												
0	0	0												
21	258	6990												
0	0	0												
0	0	0												
0	0	0												
0	0	0												
0	0	0												
0	0	0												
0	0	0												
0	0	0												
0	0	0												
0	0	0												
0	0	0												
0	0	0												
0	0	0												
0	0	0												
5051	8832	8767												
1216	2346	2287												
64	151	113												
298	339	360												
27	34	16												
2	7	10												
30	58	66												
0	0	0												
0	0	0												
0	0	0												
0	0	0												
0	0	0												

Needs to consult dengue experts on this, skip for now

Fix age, again

```
s3_2000_2016 %>% tabyl(age)
```

age	n	percent	valid_percent
-34669	1	6.760091e-06	7.947420e-06
-31868	1	6.760091e-06	7.947420e-06
-31450	1	6.760091e-06	7.947420e-06
-30781	1	6.760091e-06	7.947420e-06
-28917	1	6.760091e-06	7.947420e-06
-26725	1	6.760091e-06	7.947420e-06
-17994	1	6.760091e-06	7.947420e-06
-17993	2	1.352018e-05	1.589484e-05
-4670	1	6.760091e-06	7.947420e-06
-297	1	6.760091e-06	7.947420e-06
-2	1	6.760091e-06	7.947420e-06
0	2237	1.512232e-02	1.777838e-02
1	4871	3.292840e-02	3.871188e-02
2	2895	1.957046e-02	2.300778e-02
3	2976	2.011803e-02	2.365152e-02
4	3380	2.284911e-02	2.686228e-02
5	3941	2.664152e-02	3.132078e-02
6	4782	3.232676e-02	3.800456e-02
7	5278	3.567976e-02	4.194648e-02
8	5456	3.688306e-02	4.336112e-02
9	5494	3.713994e-02	4.366312e-02
10	5964	4.031718e-02	4.739841e-02
11	5956	4.026310e-02	4.733483e-02
12	6053	4.091883e-02	4.810573e-02
13	6165	4.167596e-02	4.899584e-02
14	5786	3.911389e-02	4.598377e-02
15	3881	2.623591e-02	3.084394e-02
16	2565	1.733963e-02	2.038513e-02
17	2435	1.646082e-02	1.935197e-02
18	2850	1.926626e-02	2.265015e-02
19	2925	1.977327e-02	2.324620e-02
20	3090	2.088868e-02	2.455753e-02
21	2946	1.991523e-02	2.341310e-02
22	2863	1.935414e-02	2.275346e-02
23	2757	1.863757e-02	2.191104e-02
24	2519	1.702867e-02	2.001955e-02

25	2241	1.514936e-02	1.781017e-02
26	2141	1.447336e-02	1.701543e-02
27	1966	1.329034e-02	1.562463e-02
28	1894	1.280361e-02	1.505241e-02
29	1720	1.162736e-02	1.366956e-02
30	1608	1.087023e-02	1.277945e-02
31	1440	9.734531e-03	1.144428e-02
32	1368	9.247805e-03	1.087207e-02
33	1190	8.044508e-03	9.457430e-03
34	967	6.537008e-03	7.685155e-03
35	931	6.293645e-03	7.399048e-03
36	857	5.793398e-03	6.810939e-03
37	770	5.205270e-03	6.119513e-03
38	670	4.529261e-03	5.324771e-03
39	569	3.846492e-03	4.522082e-03
40	531	3.589608e-03	4.220080e-03
41	484	3.271884e-03	3.846551e-03
42	397	2.683756e-03	3.155126e-03
43	377	2.548554e-03	2.996177e-03
44	302	2.041548e-03	2.400121e-03
45	295	1.994227e-03	2.344489e-03
46	270	1.825225e-03	2.145803e-03
47	240	1.622422e-03	1.907381e-03
48	215	1.453420e-03	1.708695e-03
49	183	1.237097e-03	1.454378e-03
50	213	1.439899e-03	1.692800e-03
51	204	1.379059e-03	1.621274e-03
52	140	9.464128e-04	1.112639e-03
53	127	8.585316e-04	1.009322e-03
54	121	8.179710e-04	9.616378e-04
55	109	7.368499e-04	8.662688e-04
56	105	7.098096e-04	8.344791e-04
57	74	5.002467e-04	5.881091e-04
58	116	7.841706e-04	9.219007e-04
59	73	4.934867e-04	5.801617e-04
60	66	4.461660e-04	5.245297e-04
61	55	3.718050e-04	4.371081e-04
62	63	4.258857e-04	5.006875e-04
63	51	3.447646e-04	4.053184e-04
64	50	3.380046e-04	3.973710e-04
65	37	2.501234e-04	2.940545e-04
66	27	1.825225e-04	2.145803e-04
67	29	1.960426e-04	2.304752e-04

68	28	1.892826e-04	2.225278e-04
69	26	1.757624e-04	2.066329e-04
70	33	2.230830e-04	2.622649e-04
71	23	1.554821e-04	1.827907e-04
72	23	1.554821e-04	1.827907e-04
73	17	1.149215e-04	1.351061e-04
74	15	1.014014e-04	1.192113e-04
75	12	8.112109e-05	9.536904e-05
76	13	8.788118e-05	1.033165e-04
77	6	4.056055e-05	4.768452e-05
78	8	5.408073e-05	6.357936e-05
79	9	6.084082e-05	7.152678e-05
80	21	1.419619e-04	1.668958e-04
81	13	8.788118e-05	1.033165e-04
82	8	5.408073e-05	6.357936e-05
83	4	2.704036e-05	3.178968e-05
84	5	3.380046e-05	3.973710e-05
85	3	2.028027e-05	2.384226e-05
86	2	1.352018e-05	1.589484e-05
87	3	2.028027e-05	2.384226e-05
88	3	2.028027e-05	2.384226e-05
90	25	1.690023e-04	1.986855e-04
91	5	3.380046e-05	3.973710e-05
92	5	3.380046e-05	3.973710e-05
93	1	6.760091e-06	7.947420e-06
94	1	6.760091e-06	7.947420e-06
99	1	6.760091e-06	7.947420e-06
103	1	6.760091e-06	7.947420e-06
106	1	6.760091e-06	7.947420e-06
117	1	6.760091e-06	7.947420e-06
120	1	6.760091e-06	7.947420e-06
1816	1	6.760091e-06	7.947420e-06
1825	1	6.760091e-06	7.947420e-06
1952	1	6.760091e-06	7.947420e-06
1954	1	6.760091e-06	7.947420e-06
1957	1	6.760091e-06	7.947420e-06
1958	1	6.760091e-06	7.947420e-06
1959	1	6.760091e-06	7.947420e-06
1962	3	2.028027e-05	2.384226e-05
1964	2	1.352018e-05	1.589484e-05
1970	2	1.352018e-05	1.589484e-05
1972	4	2.704036e-05	3.178968e-05
1973	1	6.760091e-06	7.947420e-06

1974	2	1.352018e-05	1.589484e-05
1975	3	2.028027e-05	2.384226e-05
1977	1	6.760091e-06	7.947420e-06
1978	2	1.352018e-05	1.589484e-05
1981	1	6.760091e-06	7.947420e-06
1982	1	6.760091e-06	7.947420e-06
1983	3	2.028027e-05	2.384226e-05
1984	2	1.352018e-05	1.589484e-05
1985	4	2.704036e-05	3.178968e-05
1986	3	2.028027e-05	2.384226e-05
1987	3	2.028027e-05	2.384226e-05
1988	2	1.352018e-05	1.589484e-05
1989	4	2.704036e-05	3.178968e-05
1990	2	1.352018e-05	1.589484e-05
1991	6	4.056055e-05	4.768452e-05
1992	4	2.704036e-05	3.178968e-05
1993	7	4.732064e-05	5.563194e-05
1994	5	3.380046e-05	3.973710e-05
1995	4	2.704036e-05	3.178968e-05
1996	4	2.704036e-05	3.178968e-05
1997	7	4.732064e-05	5.563194e-05
1998	3	2.028027e-05	2.384226e-05
1999	3	2.028027e-05	2.384226e-05
2000	5	3.380046e-05	3.973710e-05
2001	2	1.352018e-05	1.589484e-05
2002	5	3.380046e-05	3.973710e-05
2003	3	2.028027e-05	2.384226e-05
2004	2	1.352018e-05	1.589484e-05
2005	2	1.352018e-05	1.589484e-05
2006	2	1.352018e-05	1.589484e-05
2007	5	3.380046e-05	3.973710e-05
2008	5	3.380046e-05	3.973710e-05
2009	5	3.380046e-05	3.973710e-05
2010	2	1.352018e-05	1.589484e-05
2011	4	2.704036e-05	3.178968e-05
2012	4	2.704036e-05	3.178968e-05
2013	2	1.352018e-05	1.589484e-05
2014	2	1.352018e-05	1.589484e-05
2015	2	1.352018e-05	1.589484e-05
NA	22100	1.493980e-01	NA

There are a lot of YOB that are put in as age, let's quickly fix that

```
s3_2000_2016 %>%
  mutate(age = if_else(age > 1000, year(date) - age, age)) %>%
  tabyl(age)
```

age	n	percent	valid_percent
-34669	1	6.760091e-06	7.947420e-06
-31868	1	6.760091e-06	7.947420e-06
-31450	1	6.760091e-06	7.947420e-06
-30781	1	6.760091e-06	7.947420e-06
-28917	1	6.760091e-06	7.947420e-06
-26725	1	6.760091e-06	7.947420e-06
-17994	1	6.760091e-06	7.947420e-06
-17993	2	1.352018e-05	1.589484e-05
-4670	1	6.760091e-06	7.947420e-06
-297	1	6.760091e-06	7.947420e-06
-2	1	6.760091e-06	7.947420e-06
0	2246	1.518316e-02	1.784991e-02
1	4873	3.294192e-02	3.872778e-02
2	2897	1.958398e-02	2.302368e-02
3	2977	2.012479e-02	2.365947e-02
4	3381	2.285587e-02	2.687023e-02
5	3943	2.665504e-02	3.133668e-02
6	4783	3.233352e-02	3.801251e-02
7	5283	3.571356e-02	4.198622e-02
8	5461	3.691686e-02	4.340086e-02
9	5499	3.717374e-02	4.370286e-02
10	5966	4.033070e-02	4.741431e-02
11	5958	4.027662e-02	4.735073e-02
12	6055	4.093235e-02	4.812163e-02
13	6166	4.168272e-02	4.900379e-02
14	5791	3.914769e-02	4.602351e-02
15	3883	2.624943e-02	3.085983e-02
16	2570	1.737343e-02	2.042487e-02
17	2438	1.648110e-02	1.937581e-02
18	2853	1.928654e-02	2.267399e-02
19	2932	1.982059e-02	2.330184e-02
20	3095	2.092248e-02	2.459726e-02
21	2950	1.994227e-02	2.344489e-02
22	2868	1.938794e-02	2.279320e-02
23	2764	1.868489e-02	2.196667e-02
24	2524	1.706247e-02	2.005929e-02
25	2247	1.518992e-02	1.785785e-02

26	2143	1.448688e-02	1.703132e-02
27	1970	1.331738e-02	1.565642e-02
28	1896	1.281713e-02	1.506831e-02
29	1723	1.164764e-02	1.369340e-02
30	1611	1.089051e-02	1.280329e-02
31	1443	9.754811e-03	1.146813e-02
32	1370	9.261325e-03	1.088797e-02
33	1193	8.064789e-03	9.481272e-03
34	968	6.543768e-03	7.693102e-03
35	931	6.293645e-03	7.399048e-03
36	857	5.793398e-03	6.810939e-03
37	770	5.205270e-03	6.119513e-03
38	672	4.542781e-03	5.340666e-03
39	570	3.853252e-03	4.530029e-03
40	531	3.589608e-03	4.220080e-03
41	487	3.292164e-03	3.870393e-03
42	399	2.697276e-03	3.171021e-03
43	378	2.555314e-03	3.004125e-03
44	306	2.068588e-03	2.431910e-03
45	295	1.994227e-03	2.344489e-03
46	272	1.838745e-03	2.161698e-03
47	240	1.622422e-03	1.907381e-03
48	215	1.453420e-03	1.708695e-03
49	183	1.237097e-03	1.454378e-03
50	213	1.439899e-03	1.692800e-03
51	204	1.379059e-03	1.621274e-03
52	142	9.599329e-04	1.128534e-03
53	127	8.585316e-04	1.009322e-03
54	124	8.382513e-04	9.854801e-04
55	109	7.368499e-04	8.662688e-04
56	105	7.098096e-04	8.344791e-04
57	75	5.070068e-04	5.960565e-04
58	117	7.909307e-04	9.298481e-04
59	74	5.002467e-04	5.881091e-04
60	66	4.461660e-04	5.245297e-04
61	55	3.718050e-04	4.371081e-04
62	64	4.326458e-04	5.086349e-04
63	51	3.447646e-04	4.053184e-04
64	51	3.447646e-04	4.053184e-04
65	37	2.501234e-04	2.940545e-04
66	27	1.825225e-04	2.145803e-04
67	29	1.960426e-04	2.304752e-04
68	28	1.892826e-04	2.225278e-04

69	26	1.757624e-04	2.066329e-04
70	33	2.230830e-04	2.622649e-04
71	23	1.554821e-04	1.827907e-04
72	23	1.554821e-04	1.827907e-04
73	17	1.149215e-04	1.351061e-04
74	15	1.014014e-04	1.192113e-04
75	12	8.112109e-05	9.536904e-05
76	13	8.788118e-05	1.033165e-04
77	6	4.056055e-05	4.768452e-05
78	8	5.408073e-05	6.357936e-05
79	9	6.084082e-05	7.152678e-05
80	21	1.419619e-04	1.668958e-04
81	13	8.788118e-05	1.033165e-04
82	8	5.408073e-05	6.357936e-05
83	4	2.704036e-05	3.178968e-05
84	5	3.380046e-05	3.973710e-05
85	3	2.028027e-05	2.384226e-05
86	2	1.352018e-05	1.589484e-05
87	3	2.028027e-05	2.384226e-05
88	3	2.028027e-05	2.384226e-05
90	25	1.690023e-04	1.986855e-04
91	5	3.380046e-05	3.973710e-05
92	5	3.380046e-05	3.973710e-05
93	1	6.760091e-06	7.947420e-06
94	1	6.760091e-06	7.947420e-06
99	1	6.760091e-06	7.947420e-06
103	1	6.760091e-06	7.947420e-06
106	1	6.760091e-06	7.947420e-06
117	1	6.760091e-06	7.947420e-06
120	1	6.760091e-06	7.947420e-06
191	1	6.760091e-06	7.947420e-06
200	1	6.760091e-06	7.947420e-06
NA	22100	1.493980e-01	NA

Then let's drop negative age and age > 90

```
s4_2000_2016 <- s3_2000_2016 %>%
  mutate(age = if_else(age > 1000, year(date) - age, age)) %>%
  filter(age >= 0, age < 91)

s4_2000_2016 %>% skim()
```

Table 3: Data summary

Name	Piped data
Number of rows	125796
Number of columns	9
Column type frequency:	
character	7
Date	1
numeric	1
Group variables	None

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
sex	12729	0.90	1	1	0	2	0
district	0	1.00	2	10	0	24	0
commune	0	1.00	2	16	0	171	0
new_commune	8307	0.93	6	21	0	103	0
hospital	0	1.00	7	18	0	37	0
icd	5280	0.96	3	6	0	35	0
in_out_patient	0	1.00	10	10	0	1	0

Variable type: Date

skim_variable	n_missing	complete_rate	min	max	median	n_unique
date	0	1	2000-01-01	2016-12-31	2011-01-29	6128

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
age	0	1	15.95	11.67	0	8	13	22	90	

Wrap up

Probably finished cleaning 2000-2016 data for now

Let's check final number of rows

```
nrow(s4_2000_2016)
```

```
[1] 125796
```

```
start_nrow - nrow(s4_2000_2016)
```

```
[1] 22131
```

```
(start_nrow - nrow(s4_2000_2016)) / start_nrow * 100
```

```
[1] 14.96076
```

Lost about 15% of rows

Data from 2017-2025

Extract the data

```
raw_2017_2025 <- xlsx_rows[[2]] |> bind_rows()
```

Quick skim at the data

```
skim(raw_2017_2025)
```

Table 4: Data summary

Name	raw_2017_2025
Number of rows	314324
Number of columns	18
Column type frequency:	
character	9

numeric	5
POSIXct	4
Group variables	None

Variable type: character

skim_variable	n_miss- ing	com- plete_rate	min	max	empty	n_unique	whitespace
dia_chi	6849	0.98	2	249	0	276051	0
phuong_xa_cu	0	1.00	8	23	0	174	0
quan_huyen_cu	0	1.00	6	16	0	24	0
thanh_pho_cu	0	1.00	9	9	0	1	0
hinh_thuc_dieu_tri	0	1.00	7	9	0	2	0
phan_do_benh	223554	0.29	3	24	0	3	0
phan_loai_chan_doan	37749	0.88	1	79	0	142	0
diachi_api	7805	0.98	29	193	0	167617	0
px_moi	7805	0.98	6	15	0	102	0

Variable type: numeric

skim_vari- able	n_miss- ing	com- plete_rate	mean	sd	p0	p25	p50	p75	p100	hist
tuoi	48	1.00	22.41	27.45	-	10.00	20.00	32.00	2025.00	
					7974.00					
nam	0	1.00	2020.28	2.23	2017.00	2018.00	2020.00	2022.00	2025.00	
thang	0	1.00	7.41	3.37	1.00	5.00	8.00	10.00	12.00	
long	7805	0.98	106.65	0.07	106.37	106.61	106.65	106.69	106.98	
lat	7805	0.98	10.80	0.07	10.38	10.76	10.79	10.84	11.15	

Variable type: POSIXct

skim_variable	n_miss- ing	com- plete_rate	min	max	median	n_unique
ngay_thang_nam_sinh	1560	1.00	1900-06-01	9175-09-09	2000-08-18	24491
ngay_khoi_phat_trieu_chung	15799	0.95	1899-12-31	2219-06-14	2020-01-16	3158

skim_variable	n_missing	complete_rate	min	max	median	n_unique
ngay_nhap_vien_kham_benh	0	1.00	2017-01-01	2025-06-30	2020-08-01	3103
ngay_bao_cao	0	1.00	2018-01-02	2025-06-30	2020-08-04	1983

The actual columns based on our columns of interest are (with `complete_rate`):

- `age = tuoi` (0.999)
- `date of admission = ngay_nhap_vien_kham_benh` (1)
- `district = quan_huyen_cu` (1)
- `commune = phuong_xa_cu` (1)
- `new_commune = px_moi` (0.975)
- `icd = phan_do_benh` (0.252)
- `in-patient = hinh_thuc_dieu_tri` (1.00)

Selecting columns of interest

```
s1_2017_2025 <- raw_2017_2025 %>%
  select(
    tuoi,
    ngay_nhap_vien_kham_benh,
    quan_huyen_cu,
    phuong_xa_cu,
    px_moi,
    phan_do_benh,
    hinh_thuc_dieu_tri
  )
s1_2017_2025 %>% skim()
```

Table 5: Data summary

Name	Piped data
Number of rows	314324
Number of columns	7

Column type frequency:	
character	5
numeric	1
POSIXct	1
Group variables	
None	

Variable type: character

skim_variable	n_miss- ing	com- plete_rate	min	max	empty	n_unique	whitespace
quan_huyen_cu	0	1.00	6	16	0	24	0
phuong_xa_cu	0	1.00	8	23	0	174	0
px_moi	7805	0.98	6	15	0	102	0
phan_do_benh	223554	0.29	3	24	0	3	0
hinh_thuc_dieu_tri	0	1.00	7	9	0	2	0

Variable type: numeric

skim_vari- able	n_miss- ing	com- plete_rate	mean	sd	p0	p25	p50	p75	p100	hist
tuoi	48	1	22.41	27.45	- 7974	10	20	32	2025	

Variable type: POSIXct

skim_variable	n_miss- ing	com- plete_rate	min	max	median	n_unique
ngay_nhap_vien_kham_benh	0	1	2017-01-01	2025-06-30	2020-08-01	3103

Keeping track of raw number of rows at start to see how much lost during data cleaning

```
start_nrow2 <- nrow(s1_2017_2025)
start_nrow2
```

```
[1] 314324
```

Rename columns

Let's rename columns before continuing

```
s2_2017_2025 <- s1_2017_2025 %>%  
  rename(  
    age = tuoi,  
    date = ngay_nhap_vien_kham_benh,  
    district = quan_huyen_cu,  
    commune = phuong_xa_cu,  
    new_commune = px_moi,  
    icd = phan_do_benh,  
    in_out_patient = hinh_thuc_dieu_tri  
  )  
  
s2_2017_2025 %>% skim()
```

Table 6: Data summary

Name	Piped data
Number of rows	314324
Number of columns	7
Column type frequency:	
character	5
numeric	1
POSIXct	1
Group variables	None

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
district	0	1.00	6	16	0	24	0
commune	0	1.00	8	23	0	174	0
new_commune	7805	0.98	6	15	0	102	0
icd	223554	0.29	3	24	0	3	0
in_out_patient	0	1.00	7	9	0	2	0

Variable type: numeric

skim_vari- able	n_miss- ing	com- plete_rate	mean	sd	p0	p25	p50	p75	p100	hist
age	48	1	22.41	27.45	-	10	20	32	2025	
					7974					

Variable type: POSIXct

skim_vari- able	n_miss- ing	com- plete_rate	min	max	median	n_unique
date	0	1	2017-01-01	2025-06-30	2020-08-01	3103

Fix age

We see the same problem with some negative age and YOB put in as age

```
s2_2017_2025 %>% tabyl(age)
```

age	n	percent	valid_percent
-7974	1	3.181431e-06	3.181917e-06
-7153	1	3.181431e-06	3.181917e-06
-974	1	3.181431e-06	3.181917e-06
-954	1	3.181431e-06	3.181917e-06
-929	1	3.181431e-06	3.181917e-06
-77	1	3.181431e-06	3.181917e-06
-11	1	3.181431e-06	3.181917e-06
-1	1	3.181431e-06	3.181917e-06
0	1658	5.274812e-03	5.275618e-03
1	9937	3.161388e-02	3.161870e-02
2	6426	2.044387e-02	2.044700e-02
3	5873	1.868454e-02	1.868740e-02
4	5813	1.849366e-02	1.849648e-02
5	6587	2.095608e-02	2.095928e-02
6	8109	2.579822e-02	2.580216e-02
7	8501	2.704534e-02	2.704947e-02
8	8578	2.729031e-02	2.729448e-02
9	8443	2.686082e-02	2.686492e-02
10	9567	3.043675e-02	3.044140e-02
11	9549	3.037948e-02	3.038412e-02
12	9074	2.886830e-02	2.887271e-02

13	9556	3.040175e-02	3.040639e-02
14	9147	2.910055e-02	2.910499e-02
15	8382	2.666675e-02	2.667082e-02
16	7190	2.287449e-02	2.287798e-02
17	6319	2.010346e-02	2.010653e-02
18	6901	2.195505e-02	2.195841e-02
19	7974	2.536873e-02	2.537260e-02
20	7135	2.269951e-02	2.270297e-02
21	7009	2.229865e-02	2.230205e-02
22	7104	2.260088e-02	2.260434e-02
23	6937	2.206958e-02	2.207295e-02
24	6792	2.160828e-02	2.161158e-02
25	7086	2.254362e-02	2.254706e-02
26	7017	2.232410e-02	2.232751e-02
27	7040	2.239727e-02	2.240069e-02
28	6768	2.153192e-02	2.153521e-02
29	6642	2.113106e-02	2.113429e-02
30	6227	1.981077e-02	1.981379e-02
31	5879	1.870363e-02	1.870649e-02
32	5687	1.809280e-02	1.809556e-02
33	5402	1.718609e-02	1.718871e-02
34	5176	1.646708e-02	1.646960e-02
35	4972	1.581807e-02	1.582049e-02
36	4501	1.431962e-02	1.432181e-02
37	4275	1.360062e-02	1.360269e-02
38	3841	1.221988e-02	1.222174e-02
39	3763	1.197172e-02	1.197355e-02
40	3326	1.058144e-02	1.058305e-02
41	3019	9.604739e-03	9.606206e-03
42	2789	8.873010e-03	8.874365e-03
43	2596	8.258994e-03	8.260255e-03
44	2387	7.594075e-03	7.595235e-03
45	2121	6.747814e-03	6.748845e-03
46	2066	6.572836e-03	6.573840e-03
47	1824	5.802929e-03	5.803816e-03
48	1656	5.268449e-03	5.269254e-03
49	1547	4.921673e-03	4.922425e-03
50	1459	4.641707e-03	4.642416e-03
51	1357	4.317201e-03	4.317861e-03
52	1231	3.916341e-03	3.916939e-03
53	1158	3.684097e-03	3.684659e-03
54	1085	3.451852e-03	3.452379e-03
55	1015	3.229152e-03	3.229645e-03

56	946	3.009633e-03	3.010093e-03
57	920	2.926916e-03	2.927363e-03
58	856	2.723305e-03	2.723721e-03
59	766	2.436976e-03	2.437348e-03
60	755	2.401980e-03	2.402347e-03
61	674	2.144284e-03	2.144612e-03
62	644	2.048841e-03	2.049154e-03
63	562	1.787964e-03	1.788237e-03
64	554	1.762513e-03	1.762782e-03
65	438	1.393467e-03	1.393679e-03
66	460	1.463458e-03	1.463682e-03
67	388	1.234395e-03	1.234584e-03
68	358	1.138952e-03	1.139126e-03
69	295	9.385220e-04	9.386654e-04
70	270	8.589863e-04	8.591175e-04
71	242	7.699062e-04	7.700238e-04
72	180	5.726575e-04	5.727450e-04
73	192	6.108347e-04	6.109280e-04
74	148	4.708517e-04	4.709236e-04
75	139	4.422189e-04	4.422864e-04
76	111	3.531388e-04	3.531927e-04
77	93	2.958730e-04	2.959182e-04
78	90	2.863288e-04	2.863725e-04
79	85	2.704216e-04	2.704629e-04
80	66	2.099744e-04	2.100065e-04
81	76	2.417887e-04	2.418257e-04
82	66	2.099744e-04	2.100065e-04
83	54	1.717973e-04	1.718235e-04
84	38	1.208944e-04	1.209128e-04
85	40	1.272572e-04	1.272767e-04
86	35	1.113501e-04	1.113671e-04
87	30	9.544292e-05	9.545750e-05
88	32	1.018058e-04	1.018213e-04
89	29	9.226149e-05	9.227558e-05
90	23	7.317290e-05	7.318408e-05
91	16	5.090289e-05	5.091066e-05
92	17	5.408432e-05	5.409258e-05
93	8	2.545145e-05	2.545533e-05
94	7	2.227001e-05	2.227342e-05
95	5	1.590715e-05	1.590958e-05
96	4	1.272572e-05	1.272767e-05
98	2	6.362861e-06	6.363833e-06
99	1	3.181431e-06	3.181917e-06

101	2	6.362861e-06	6.363833e-06
103	1	3.181431e-06	3.181917e-06
106	1	3.181431e-06	3.181917e-06
110	1	3.181431e-06	3.181917e-06
112	8	2.545145e-05	2.545533e-05
113	4	1.272572e-05	1.272767e-05
114	6	1.908858e-05	1.909150e-05
117	4	1.272572e-05	1.272767e-05
122	4	1.272572e-05	1.272767e-05
145	1	3.181431e-06	3.181917e-06
147	1	3.181431e-06	3.181917e-06
195	1	3.181431e-06	3.181917e-06
197	2	6.362861e-06	6.363833e-06
198	6	1.908858e-05	1.909150e-05
199	4	1.272572e-05	1.272767e-05
200	1	3.181431e-06	3.181917e-06
223	1	3.181431e-06	3.181917e-06
232	2	6.362861e-06	6.363833e-06
345	1	3.181431e-06	3.181917e-06
352	1	3.181431e-06	3.181917e-06
427	1	3.181431e-06	3.181917e-06
440	1	3.181431e-06	3.181917e-06
551	1	3.181431e-06	3.181917e-06
819	1	3.181431e-06	3.181917e-06
820	1	3.181431e-06	3.181917e-06
821	4	1.272572e-05	1.272767e-05
822	2	6.362861e-06	6.363833e-06
823	1	3.181431e-06	3.181917e-06
824	2	6.362861e-06	6.363833e-06
825	2	6.362861e-06	6.363833e-06
925	1	3.181431e-06	3.181917e-06
1010	1	3.181431e-06	3.181917e-06
1013	1	3.181431e-06	3.181917e-06
1038	1	3.181431e-06	3.181917e-06
2013	1	3.181431e-06	3.181917e-06
2016	1	3.181431e-06	3.181917e-06
2019	2	6.362861e-06	6.363833e-06
2021	1	3.181431e-06	3.181917e-06
2023	1	3.181431e-06	3.181917e-06
2025	2	6.362861e-06	6.363833e-06
NA	48	1.527087e-04	NA

```
s3_2017_2025 <- s2_2017_2025 %>%
  mutate(age = if_else(age > 2000, year(date) - age, age)) %>%
  filter(age >= 0, age < 91)

s3_2017_2025 %>% tabyl(age)
```

age	n	percent
0	1661	5.287519e-03
1	9938	3.163598e-02
2	6426	2.045611e-02
3	5873	1.869572e-02
4	5815	1.851109e-02
5	6587	2.096863e-02
6	8109	2.581366e-02
7	8502	2.706471e-02
8	8578	2.730664e-02
9	8444	2.688008e-02
10	9567	3.045496e-02
11	9549	3.039766e-02
12	9074	2.888558e-02
13	9556	3.041995e-02
14	9147	2.911796e-02
15	8382	2.668271e-02
16	7190	2.288818e-02
17	6319	2.011549e-02
18	6901	2.196819e-02
19	7974	2.538391e-02
20	7135	2.271309e-02
21	7009	2.231199e-02
22	7104	2.261441e-02
23	6937	2.208279e-02
24	6792	2.162121e-02
25	7086	2.255711e-02
26	7017	2.233746e-02
27	7040	2.241068e-02
28	6768	2.154481e-02
29	6642	2.114371e-02
30	6227	1.982262e-02
31	5879	1.871482e-02
32	5687	1.810362e-02
33	5402	1.719637e-02
34	5176	1.647694e-02

35 4972 1.582754e-02
36 4501 1.432819e-02
37 4275 1.360876e-02
38 3841 1.222719e-02
39 3763 1.197889e-02
40 3326 1.058777e-02
41 3019 9.610487e-03
42 2789 8.878320e-03
43 2596 8.263937e-03
44 2387 7.598620e-03
45 2121 6.751853e-03
46 2066 6.576769e-03
47 1824 5.806402e-03
48 1656 5.271602e-03
49 1547 4.924619e-03
50 1459 4.644485e-03
51 1357 4.319785e-03
52 1231 3.918685e-03
53 1158 3.686301e-03
54 1085 3.453918e-03
55 1015 3.231085e-03
56 946 3.011435e-03
57 920 2.928668e-03
58 856 2.724934e-03
59 766 2.438434e-03
60 755 2.403418e-03
61 674 2.145568e-03
62 644 2.050067e-03
63 562 1.789034e-03
64 554 1.763567e-03
65 438 1.394301e-03
66 460 1.464334e-03
67 388 1.235134e-03
68 358 1.139634e-03
69 295 9.390837e-04
70 270 8.595003e-04
71 242 7.703670e-04
72 180 5.730002e-04
73 192 6.112002e-04
74 148 4.711335e-04
75 139 4.424835e-04
76 111 3.533501e-04
77 93 2.960501e-04

```

78  90 2.865001e-04
79  85 2.705834e-04
80  66 2.101001e-04
81  76 2.419334e-04
82  66 2.101001e-04
83  54 1.719001e-04
84  38 1.209667e-04
85  40 1.273334e-04
86  35 1.114167e-04
87  30 9.550004e-05
88  32 1.018667e-04
89  29 9.231670e-05
90  23 7.321670e-05

```

```
s3_2017_2025 %>% skim()
```

Table 7: Data summary

Name	Piped data
Number of rows	314136
Number of columns	7
Column type frequency:	
character	5
numeric	1
POSIXct	1
Group variables	None

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
district	0	1.00	6	16	0	24	0
commune	0	1.00	8	23	0	174	0
new_commune	7801	0.98	6	15	0	102	0
icd	223431	0.29	3	24	0	3	0
in_out_patient	0	1.00	7	9	0	2	0

Variable type: numeric

skim_vari- able	n_miss- ing	com- plete_rate	mean	sd	p0	p25	p50	p75	p100	hist
age	0	1	22.33	15.19	0	10	20	32	90	

Variable type: POSIXct

skim_vari- able	n_miss- ing	com- plete_rate	min	max	median	n_unique
date	0	1	2017-01-01	2025-06-30	2020-07-31	3103

Fix dates

Date data is in `datetime`, convert to `date` only

```
s4_2017_2025 <- s3_2017_2025 %>%
  mutate(date = convert_to_date(date))

s4_2017_2025 %>% skim()
```

Table 8: Data summary

Name	Piped data
Number of rows	314136
Number of columns	7
Column type frequency:	
character	5
Date	1
numeric	1
Group variables	None

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
district	0	1.00	6	16	0	24	0
commune	0	1.00	8	23	0	174	0
new_commune	7801	0.98	6	15	0	102	0

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
icd	223431	0.29	3	24	0	3	0
in_out_patient	0	1.00	7	9	0	2	0

Variable type: Date

skim_variable	n_missing	complete_rate	min	max	median	n_unique
date	0	1	2017-01-01	2025-06-30	2020-07-31	3103

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
age	0	1	22.33	15.19	0	10	20	32	90	

Fix ICD

```
s4_2017_2025 %>%
  mutate(year = year(date), icd = tolower(icd)) %>%
  tabyl(icd, year)
```

```

      icd 2017 2018 2019 2020 2021 2022 2023 2024 2025
      sxh 2925 7701 13386 5767 3380 22179 5942 5380 2183
sxh có dấu hiệu cảnh báo 706 606 697 596 536 9386 2324 2057 991
      sxh nặng 261 157 122 110 99 2101 514 451 148
      <NA> 29827 37480 54303 19463 8660 48144 10324 8459 6771
```

Mostly inconsistent letter casing, the classes are rather consistent

```
s5_2017_2025 <- s4_2017_2025 %>%
  mutate(icd = tolower(icd))

s5_2017_2025 %>%
  mutate(year = year(date)) %>%
  tabyl(icd, year)
```

	icd	2017	2018	2019	2020	2021	2022	2023	2024	2025
	sxh	2925	7701	13386	5767	3380	22179	5942	5380	2183
sxh có dấu hiệu cảnh báo		706	606	697	596	536	9386	2324	2057	991
sxh nặng		261	157	122	110	99	2101	514	451	148
<NA>		29827	37480	54303	19463	8660	48144	10324	8459	6771

Fix districts

```
s6_2017_2025 <- s5_2017_2025 %>% mutate(district = tolower(district))
s6_2017_2025 %>% skim()
```

Table 9: Data summary

Name	Piped data
Number of rows	314136
Number of columns	7
Column type frequency:	
character	5
Date	1
numeric	1
Group variables	None

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
district	0	1.00	6	16	0	24	0
commune	0	1.00	8	23	0	174	0
new_commune	7801	0.98	6	15	0	102	0
icd	223431	0.29	3	24	0	3	0
in_out_patient	0	1.00	7	9	0	2	0

Variable type: Date

skim_vari- able	n_miss- ing	com- plete_rate	min	max	median	n_unique
date	0	1	2017-01-01	2025-06-30	2020-07-31	3103

Variable type: numeric

skim_vari- able	n_miss- ing	com- plete_rate	mean	sd	p0	p25	p50	p75	p100	hist
age	0	1	22.33	15.19	0	10	20	32	90	

Fix in-/out-patient

Now let's fix in-patient and out-patient classification

```
s6_2017_2025 %>% tabyl(in_out_patient)
```

```
in_out_patient      n    percent
    Ngoại trú 136509 0.4345538
    Nội trú   177627 0.5654462
```

Simplest things to do now are remove diacritics, normalise casing, normalizing spacing

```
s7_2017_2025 <- s6_2017_2025 %>%
  mutate(
    in_out_patient = stri_trans_general(in_out_patient, id = "Latin-ASCII") %>%
      tolower()
  )

s7_2017_2025 %>%
  tabyl(in_out_patient) %>%
  arrange(desc(n))
```

```
in_out_patient      n    percent
    noi tru 177627 0.5654462
    ngoai tru 136509 0.4345538
```

Relabel in_out_patient status

```
s8_2017_2025 <- s7_2017_2025 %>%
  mutate(
    in_out_patient = case_when(
      in_out_patient == "noi tru" ~ "in-patient",
      in_out_patient == "ngoai tru" ~ "out-patient",
      .default = "other"
    )
  )
```

```
s8_2017_2025 %>%
  tabyl(in_out_patient)
```

```
in_out_patient      n    percent
in-patient 177627 0.5654462
out-patient 136509 0.4345538
```

```
s8_2017_2025 %>% skim()
```

Table 10: Data summary

Name	Piped data
Number of rows	314136
Number of columns	7
Column type frequency:	
character	5
Date	1
numeric	1
Group variables	None

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
district	0	1.00	6	16	0	24	0
commune	0	1.00	8	23	0	174	0
new_commune	7801	0.98	6	15	0	102	0
icd	223431	0.29	3	24	0	3	0
in_out_patient	0	1.00	10	11	0	2	0

Variable type: Date

skim_variable	n_missing	complete_rate	min	max	median	n_unique
date	0	1	2017-01-01	2025-06-30	2020-07-31	3103

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
age	0	1	22.33	15.19	0	10	20	32	90	

Wrap up

Probably finished cleaning 2017-2025 data

Let's check final number of rows

```
nrow(s8_2017_2025)
```

```
[1] 314136
```

```
start_nrow2 - nrow(s8_2017_2025)
```

```
[1] 188
```

```
(start_nrow2 - nrow(s8_2017_2025)) / start_nrow2 * 100
```

```
[1] 0.0598109
```

Lost less than 1% of rows

Data after the 2025 reform

Extract the data

```
raw_aug2025 <- xlsx_raws[[3]] |> bind_rows()
```

Quick skim at the data

```
skim(raw_aug2025)
```

Table 11: Data summary

Name	raw_aug2025
Number of rows	49673
Number of columns	18
Column type frequency:	
character	9
numeric	5
POSIXct	4
Group variables	None

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
dia_chi	1221	0.98	2	177	0	43839	0
phuong_xa_moi	0	1.00	9	23	0	168	0
hinh_thuc_dieu_tri	0	1.00	7	9	0	2	0
phan_do_benh	0	1.00	3	24	0	3	0
phan_loai_chan_doan	0	1.00	6	25	0	3	0
diachi_api	9724	0.80	31	153	0	30065	0
phuong_xa_cu	9733	0.80	8	23	0	323	0
quan_huyen_cu	9724	0.80	6	21	0	38	0
thanh_pho_cu	9724	0.80	15	21	0	3	0

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
tuoi	0	1.0	23.62	15.75	0.00	11.00	21.00	34.00	101.00	
nam	0	1.0	2025.00	0.00	2025.00	2025.00	2025.00	2025.00	2025.00	

skim_vari- able	n_miss- ing	com- plete_rate	mean	sd	p0	p25	p50	p75	p100	hist
thang	0	1.0	9.98	1.43	8.00	9.00	10.00	11.00	12.00	
long	9724	0.8	106.70	0.15	106.34	106.62	106.67	106.74	107.57	
lat	9724	0.8	10.84	0.16	8.68	10.76	10.82	10.92	11.47	

Variable type: POSIXct

skim_variable	n_miss- ing	com- plete_rate	min	max	median	n_unique
ngay_thang_nam_sinh	0	1	1924-12-01	2025-12-28	2004-12-05	17343
ngay_khoi_phat_trieu_chung	0	1	1981-10-24	2025-12-31	2025-10-13	205
ngay_nhap_vien_kham_benh	0	1	2025-08-01	2025-12-31	2025-10-15	153
ngay_bao_cao	0	1	2025-08-01	2026-02-11	2025-10-17	169

The actual columns based on our columns of interest are (with `complete_rate`):

- `age = tuoi` (1)
- `date of admission = ngay_nhap_vien_kham_benh` (1)
- `district = quan_huyen_cu` (0.804)
- `commune = phuong_xa_cu` (0.804)
- `new_commune = phuong_xa_moi` (1)
- `icd = phan_do_benh` (1)
- `in-patient = hinh_thuc_dieu_tri` (1.00)

Selecting columns of interest

```
s1_aug2025 <- raw_aug2025 %>%
  select(
    tuoi,
    ngay_nhap_vien_kham_benh,
    quan_huyen_cu,
```

```

    phuong_xa_cu,
    phuong_xa_moi,
    phan_do_benh,
    hinh_thuc_dieu_tri
  )
s1_aug2025 %>% skim()

```

Table 12: Data summary

Name	Piped data
Number of rows	49673
Number of columns	7
Column type frequency:	
character	5
numeric	1
POSIXct	1
Group variables	None

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
quan_huyen_cu	9724	0.8	6	21	0	38	0
phuong_xa_cu	9733	0.8	8	23	0	323	0
phuong_xa_moi	0	1.0	9	23	0	168	0
phan_do_benh	0	1.0	3	24	0	3	0
hinh_thuc_dieu_tri	0	1.0	7	9	0	2	0

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
tuoi	0	1	23.62	15.75	0	11	21	34	101	

Variable type: POSIXct

skim_variable	n_missing	complete_rate	min	max	median	n_unique
ngay_nhap_vien_kham_benh	0	1	2025-08-01	2025-12-31	2025-10-15	153

Keeping track of raw number of rows at start to see how much lost during data cleaning

```
start_nrow3 <- nrow(s1_aug2025)
start_nrow3
```

```
[1] 49673
```

Rename columns

Let's rename columns before continuing

```
s2_aug2025 <- s1_aug2025 %>%
  rename(
    age = tuoi,
    date = ngay_nhap_vien_kham_benh,
    district = quan_huyen_cu,
    commune = phuong_xa_cu,
    new_commune = phuong_xa_moi,
    icd = phan_do_benh,
    in_out_patient = hinh_thuc_dieu_tri
  )

s2_aug2025 %>% skim()
```

Table 13: Data summary

Name	Piped data
Number of rows	49673
Number of columns	7
Column type frequency:	
character	5
numeric	1
POSIXct	1

Group variables	None
-----------------	------

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
district	9724	0.8	6	21	0	38	0
commune	9733	0.8	8	23	0	323	0
new_commune	0	1.0	9	23	0	168	0
icd	0	1.0	3	24	0	3	0
in_out_patient	0	1.0	7	9	0	2	0

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
age	0	1	23.62	15.75	0	11	21	34	101	

Variable type: POSIXct

skim_variable	n_missing	complete_rate	min	max	median	n_unique
date	0	1	2025-08-01	2025-12-31	2025-10-15	153

Fix age

Age looks quite good in this case, no YoB in data and no negative value

However, filter out cases where age > 91 to be consistent

```
s2_aug2025 %>% tabyl(age)
```

age	n	percent
0	391	7.871479e-03
1	1145	2.305075e-02
2	852	1.715218e-02
3	615	1.238097e-02

4	787	1.584362e-02
5	817	1.644757e-02
6	1186	2.387615e-02
7	1167	2.349365e-02
8	1340	2.697643e-02
9	1283	2.582892e-02
10	1466	2.951302e-02
11	1547	3.114368e-02
12	1597	3.215026e-02
13	1875	3.774686e-02
14	1511	3.041894e-02
15	1075	2.164154e-02
16	1277	2.570813e-02
17	1145	2.305075e-02
18	1355	2.727840e-02
19	1162	2.339299e-02
20	1176	2.367483e-02
21	1134	2.282930e-02
22	1113	2.240654e-02
23	1039	2.091680e-02
24	946	1.904455e-02
25	985	1.982969e-02
26	822	1.654823e-02
27	777	1.564230e-02
28	918	1.848086e-02
29	863	1.737362e-02
30	879	1.769573e-02
31	916	1.844060e-02
32	977	1.966863e-02
33	878	1.767560e-02
34	780	1.570270e-02
35	917	1.846073e-02
36	783	1.576309e-02
37	763	1.536046e-02
38	692	1.393111e-02
39	631	1.270308e-02
40	626	1.260242e-02
41	667	1.342782e-02
42	577	1.161597e-02
43	508	1.022688e-02
44	441	8.878063e-03
45	407	8.193586e-03
46	404	8.133191e-03

47	361	7.267530e-03
48	305	6.140157e-03
49	338	6.804501e-03
50	309	6.220683e-03
51	249	5.012784e-03
52	238	4.791335e-03
53	253	5.093310e-03
54	183	3.684094e-03
55	188	3.784752e-03
56	182	3.663962e-03
57	163	3.281461e-03
58	149	2.999617e-03
59	150	3.019749e-03
60	129	2.596984e-03
61	126	2.536589e-03
62	115	2.315141e-03
63	100	2.013166e-03
64	74	1.489743e-03
65	105	2.113824e-03
66	91	1.831981e-03
67	88	1.771586e-03
68	80	1.610533e-03
69	73	1.469611e-03
70	78	1.570270e-03
71	64	1.288426e-03
72	24	4.831599e-04
73	41	8.253981e-04
74	21	4.227649e-04
75	31	6.240815e-04
76	26	5.234232e-04
77	17	3.422382e-04
78	16	3.221066e-04
79	17	3.422382e-04
80	7	1.409216e-04
81	11	2.214483e-04
82	8	1.610533e-04
83	8	1.610533e-04
84	2	4.026332e-05
85	9	1.811849e-04
86	5	1.006583e-04
87	5	1.006583e-04
88	6	1.207900e-04
89	3	6.039498e-05

91	5	1.006583e-04
92	4	8.052664e-05
93	1	2.013166e-05
95	2	4.026332e-05
101	1	2.013166e-05

```
s3_aug2025 <- s2_aug2025 %>%
  filter(age >= 0, age < 91)

s3_aug2025 %>% tabyl(age)
```

age	n	percent
0	391	7.873540e-03
1	1145	2.305679e-02
2	852	1.715667e-02
3	615	1.238421e-02
4	787	1.584776e-02
5	817	1.645187e-02
6	1186	2.388240e-02
7	1167	2.349980e-02
8	1340	2.698349e-02
9	1283	2.583568e-02
10	1466	2.952074e-02
11	1547	3.115183e-02
12	1597	3.215868e-02
13	1875	3.775675e-02
14	1511	3.042690e-02
15	1075	2.164720e-02
16	1277	2.571486e-02
17	1145	2.305679e-02
18	1355	2.728554e-02
19	1162	2.339911e-02
20	1176	2.368103e-02
21	1134	2.283528e-02
22	1113	2.241240e-02
23	1039	2.092227e-02
24	946	1.904954e-02
25	985	1.983488e-02
26	822	1.655256e-02
27	777	1.564640e-02
28	918	1.848570e-02
29	863	1.737817e-02

30	879	1.770036e-02
31	916	1.844543e-02
32	977	1.967378e-02
33	878	1.768023e-02
34	780	1.570681e-02
35	917	1.846557e-02
36	783	1.576722e-02
37	763	1.536448e-02
38	692	1.393476e-02
39	631	1.270640e-02
40	626	1.260572e-02
41	667	1.343133e-02
42	577	1.161901e-02
43	508	1.022956e-02
44	441	8.880387e-03
45	407	8.195731e-03
46	404	8.135320e-03
47	361	7.269432e-03
48	305	6.141764e-03
49	338	6.806283e-03
50	309	6.222312e-03
51	249	5.014096e-03
52	238	4.792590e-03
53	253	5.094644e-03
54	183	3.685058e-03
55	188	3.785743e-03
56	182	3.664921e-03
57	163	3.282320e-03
58	149	3.000403e-03
59	150	3.020540e-03
60	129	2.597664e-03
61	126	2.537253e-03
62	115	2.315747e-03
63	100	2.013693e-03
64	74	1.490133e-03
65	105	2.114378e-03
66	91	1.832461e-03
67	88	1.772050e-03
68	80	1.610954e-03
69	73	1.469996e-03
70	78	1.570681e-03
71	64	1.288764e-03
72	24	4.832863e-04

```

73  41 8.256142e-04
74  21 4.228756e-04
75  31 6.242449e-04
76  26 5.235602e-04
77  17 3.423278e-04
78  16 3.221909e-04
79  17 3.423278e-04
80   7 1.409585e-04
81  11 2.215062e-04
82   8 1.610954e-04
83   8 1.610954e-04
84   2 4.027386e-05
85   9 1.812324e-04
86   5 1.006847e-04
87   5 1.006847e-04
88   6 1.208216e-04
89   3 6.041079e-05

```

```
s3_aug2025 %>% skim()
```

Table 14: Data summary

Name	Piped data
Number of rows	49660
Number of columns	7
Column type frequency:	
character	5
numeric	1
POSIXct	1
Group variables	None

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
district	9723	0.8	6	21	0	38	0
commune	9732	0.8	8	23	0	323	0
new_commune	0	1.0	9	23	0	168	0
icd	0	1.0	3	24	0	3	0

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
in_out_patient	0	1.0	7	9	0	2	0

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
age	0	1	23.6	15.71	0	11	21	34	89	

Variable type: POSIXct

skim_variable	n_missing	complete_rate	min	max	median	n_unique
date	0	1	2025-08-01	2025-12-31	2025-10-15	153

Fix dates

Date data is in `datetime`, convert to `date` only

```
s4_aug2025 <- s3_aug2025 %>%
  mutate(date = convert_to_date(date))

s4_aug2025 %>% skim()
```

Table 15: Data summary

Name	Piped data
Number of rows	49660
Number of columns	7
Column type frequency:	
character	5
Date	1
numeric	1
Group variables	None

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
district	9723	0.8	6	21	0	38	0
commune	9732	0.8	8	23	0	323	0
new_commune	0	1.0	9	23	0	168	0
icd	0	1.0	3	24	0	3	0
in_out_patient	0	1.0	7	9	0	2	0

Variable type: Date

skim_variable	n_missing	complete_rate	min	max	median	n_unique
date	0	1	2025-08-01	2025-12-31	2025-10-15	153

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
age	0	1	23.6	15.71	0	11	21	34	89	

Fix ICD

```
s4_aug2025 %>%
  mutate(year = year(date), icd = tolower(icd)) %>%
  tabyl(icd, year)
```

```

              icd 2025
              sxh 39042
sxh có dấu hiệu cảnh báo 8774
              sxh nặng 1844
```

Mostly inconsistent letter casing, the classes are rather consistent

```
s5_aug2025 <- s4_aug2025 %>%
  mutate(icd = tolower(icd))

s5_aug2025 %>%
  mutate(year = year(date)) %>%
  tabyl(icd, year)
```

```

            icd 2025
            sxh 39042
sxh có dấu hiệu cảnh báo 8774
            sxh nặng 1844

```

Fix districts

```

s6_aug2025 <- s5_aug2025 %>% mutate(district = tolower(district))

s6_aug2025 %>% skim()

```

Table 16: Data summary

Name	Piped data
Number of rows	49660
Number of columns	7
Column type frequency:	
character	5
Date	1
numeric	1
Group variables	None

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
district	9723	0.8	6	21	0	38	0
commune	9732	0.8	8	23	0	323	0
new__commune	0	1.0	9	23	0	168	0
icd	0	1.0	3	24	0	3	0
in_out_patient	0	1.0	7	9	0	2	0

Variable type: Date

skim_vari- able	n_miss- ing	com- plete_rate	min	max	median	n_unique
date	0	1	2025-08-01	2025-12-31	2025-10-15	153

Variable type: numeric

skim_vari- able	n_miss- ing	com- plete_rate	mean	sd	p0	p25	p50	p75	p100	hist
age	0	1	23.6	15.71	0	11	21	34	89	

Fix in-/out-patient

Now let's fix in-patient and out-patient classification

```
s6_aug2025 %>% tabyl(in_out_patient)
```

```
in_out_patient    n    percent
    Ngoại trú 20057 0.4038864
    Nội trú 29603 0.5961136
```

```
s7_aug2025 <- s6_aug2025 %>%
  mutate(
    in_out_patient = stri_trans_general(in_out_patient, id = "Latin-ASCII") %>%
      tolower()
  )

s7_aug2025 %>%
  tabyl(in_out_patient) %>%
  arrange(desc(n))
```

```
in_out_patient    n    percent
    noi tru 29603 0.5961136
    ngoai tru 20057 0.4038864
```

Relabel in_out_patient status

```
s8_aug2025 <- s7_aug2025 %>%
  mutate(
    in_out_patient = case_when(
      in_out_patient == "noi tru" ~ "in-patient",
      in_out_patient == "ngoai tru" ~ "out-patient",
      .default = "other"
    )
  )
```

```
s8_aug2025 %>%
  tabyl(in_out_patient)
```

```
in_out_patient    n    percent
in-patient 29603 0.5961136
out-patient 20057 0.4038864
```

```
s8_aug2025 %>% skim()
```

Table 17: Data summary

Name	Piped data
Number of rows	49660
Number of columns	7
Column type frequency:	
character	5
Date	1
numeric	1
Group variables	None

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
district	9723	0.8	6	21	0	38	0
commune	9732	0.8	8	23	0	323	0
new_commune	0	1.0	9	23	0	168	0
icd	0	1.0	3	24	0	3	0
in_out_patient	0	1.0	10	11	0	2	0

Variable type: Date

skim_vari- able	n_miss- ing	com- plete_rate	min	max	median	n_unique
date	0	1	2025-08-01	2025-12-31	2025-10-15	153

Variable type: numeric

skim_vari- able	n_miss- ing	com- plete_rate	mean	sd	p0	p25	p50	p75	p100	hist
age	0	1	23.6	15.71	0	11	21	34	89	

Wrap up

Check final number of rows

```
nrow(s8_aug2025)
```

```
[1] 49660
```

```
start_nrow3 - nrow(s8_aug2025)
```

```
[1] 13
```

```
(start_nrow3 - nrow(s8_aug2025)) / start_nrow3 * 100
```

```
[1] 0.02617116
```

Data joining

For now, we only need incidence from 2000-2020 with the label for the corresponding new commune

```
cleaned_incidence_dat <- s4_2000_2016 |>  
  bind_rows(s8_2017_2025) |>  
  filter(date < as.Date("2020-01-01"))  
cleaned_incidence_dat %>% skim()
```

Table 18: Data summary

Name	Piped data
Number of rows	273967
Number of columns	9
Column type frequency:	
character	7
Date	1
numeric	1
Group variables	None

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
sex	160900	0.41	1	1	0	2	0
district	0	1.00	2	16	0	48	0
commune	0	1.00	2	23	0	345	0
new_commune	12095	0.96	6	21	0	103	0
hospital	148171	0.46	7	18	0	37	0
icd	126890	0.54	3	24	0	38	0
in_out_patient	0	1.00	10	11	0	2	0

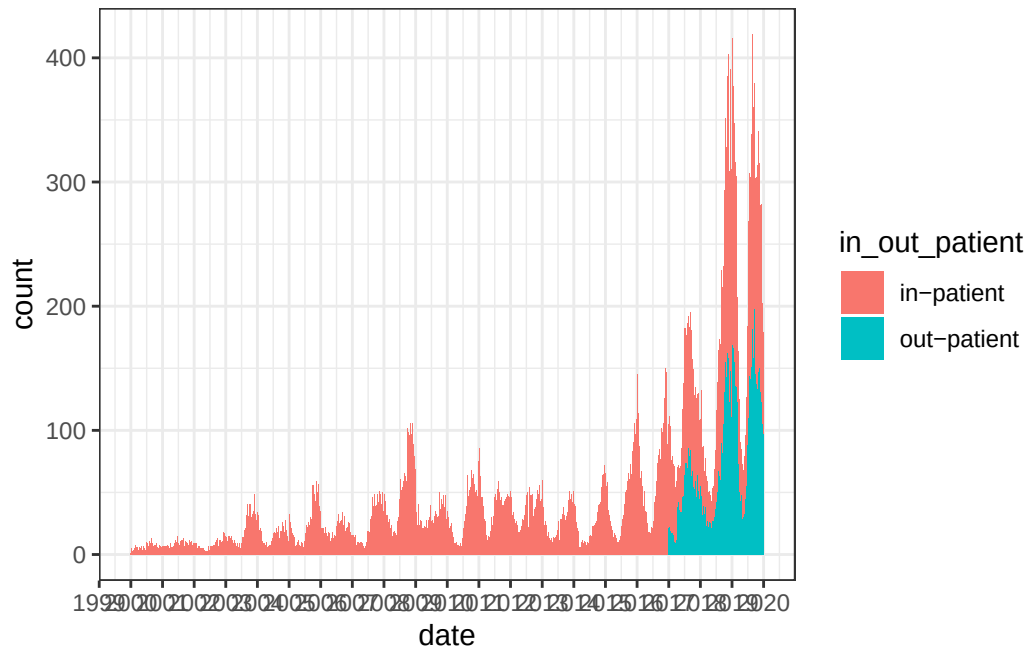
Variable type: Date

skim_variable	n_missing	complete_rate	min	max	median	n_unique
date	0	1	2000-01-01	2019-12-31	2017-06-29	7223

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
age	0	1	19.69	13.91	0	9	17	28	90	

```
cleaned_incidence_dat %>%
  ggplot() +
  geom_bar(aes(x = date, group = in_out_patient, fill = in_out_patient)) +
  scale_x_date(date_breaks = "1 year", date_labels = "%Y")
```



Normalise district and commune

Let's normalise district names from the 2 different reporting systems

```
cleaned_incidence_dat %>%
  tabyl(district)
```

district	n	percent
01	4479	0.016348684
02	2201	0.008033814
03	3905	0.014253541
04	3333	0.012165699
05	4435	0.016188081
06	7584	0.027682166
07	6352	0.023185274
08	12223	0.044614862

09	3205	0.011698489
10	5344	0.019505999
11	5848	0.021345637
12	5191	0.018947537
BINH CHANH	7801	0.028474232
BINH TAN	8820	0.032193658
BINH THANH	6418	0.023426179
CAN GIO	994	0.003628174
CU CHI	2030	0.007409652
GO VAP	4916	0.017943767
HOC MON	3994	0.014578398
NHA BE	2608	0.009519395
PHU NHUAN	2182	0.007964463
TAN BINH	9393	0.034285151
TAN PHU	6949	0.025364369
THU DUC	5591	0.020407567
huyện bình chánh	10789	0.039380655
huyện cần giờ	942	0.003438370
huyện củ chi	5694	0.020783525
huyện hóc môn	9119	0.033285031
huyện nhà bè	2998	0.010942924
quận 1	3877	0.014151339
quận 10	3463	0.012640208
quận 11	3890	0.014198790
quận 12	10425	0.038052028
quận 2	3222	0.011760541
quận 3	3076	0.011227630
quận 4	2749	0.010034055
quận 5	2812	0.010264010
quận 6	3982	0.014534597
quận 7	5401	0.019714053
quận 8	7325	0.026736797
quận 9	4497	0.016414386
quận bình thành	8314	0.030346721
quận bình tân	17467	0.063755854
quận gò vấp	5194	0.018958488
quận phú nhuận	2641	0.009639847
quận thủ đức	7364	0.026879150
quận tân bình	9592	0.035011516
quận tân phú	13338	0.048684696

```
s2_cleaned_incidence_dat <- cleaned_incidence_dat %>%
  mutate(
    district = stri_trans_general(district, id = "Latin-ASCII") %>% tolower()
  ) %>%
  mutate(district = str_remove(district, "huyen|quan") %>% trimws()) %>%
  mutate(district = trimws(district, which = "left", whitespace = "0"))

s2_cleaned_incidence_dat %>%
  tabyl(district) |>
  dim()
```

[1] 24 3

Now normalise commune names

```
s2_cleaned_incidence_dat %>%
  tabyl(district, commune)
```

district	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
10	326	318	146	250	167	147	128	194	332	228	234	461	389	625	452	0
11	327	222	391	142	622	196	248	299	193	203	320	209	291	352	236	249
12	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
3	283	231	241	311	275	160	199	223	246	145	373	112	96	243	0	0
4	147	126	178	236	92	147	0	169	126	116	0	126	128	181	180	227
5	319	261	174	230	282	106	175	147	215	156	189	95	188	214	151	0
6	231	203	187	285	255	307	219	382	221	440	512	538	419	379	0	0
7	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
8	462	479	477	828	841	575	538	268	394	326	186	274	138	299	511	471
9	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0
binh chanh	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0
binh tan	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
binh thanh	165	184	209	0	121	81	131	0	0	0	290	324	336	84	292	0
can gio	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
cu chi	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
go vap	213	0	315	140	394	100	253	149	101	331	362	619	186	176	171	380
hoc mon	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
nha be	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
phu nhuan	142	178	73	98	132	0	205	102	183	107	80	79	106	91	170	0
tan binh	217	431	233	520	308	381	276	512	586	1007	656	625	740	566	1045	47

tan phu			0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
thu duc			0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
17	18	19	21	22	24	25	26	27	28	AN	KHANH	AN	LAC	AN	LAC	A	AN	LOI	DONG
0	0	0	0	0	0	0	0	0	0		0		0		0				0
0	0	0	0	0	0	0	0	0	0		0		0		0				0
0	0	0	0	0	0	0	0	0	0		0		0		0				0
0	0	0	0	0	0	0	0	0	0		0		0		0				0
0	0	0	0	0	0	0	0	0	0		136		0		0				31
0	0	0	0	0	0	0	0	0	0		0		0		0				0
0	116	0	0	0	0	0	0	0	0		0		0		0				0
0	0	0	0	0	0	0	0	0	0		0		0		0				0
0	0	0	0	0	0	0	0	0	0		0		0		0				0
0	0	0	0	0	0	0	0	0	0		0		0		0				0
0	0	0	0	0	0	0	0	0	0		0		0		0				0
0	0	0	0	0	0	0	0	0	0		0		0		0				0
0	0	0	0	0	0	0	0	0	0		0		0		0				0
1	0	0	0	0	0	0	0	0	0		0		822		247				0
299	0	199	288	445	274	608	533	270	103		0		0		0				0
0	0	0	0	0	0	0	0	1	0		0		0		0				0
0	0	0	0	0	0	0	0	0	0		0		0		0				0
327	0	0	0	0	0	0	0	0	0		0		0		0				0
0	0	0	0	0	0	0	0	0	0		0		0		0				0
0	0	0	0	0	0	0	0	0	0		0		0		0				0
136	0	0	0	0	0	0	0	0	0		0		0		0				0
55	142	0	0	0	0	0	0	0	0		0		0		0				0
0	0	0	0	0	0	0	0	0	0		0		1		0				0
0	0	0	0	0	0	0	0	0	0		0		0		0				0
AN NHON TAY			AN PHU AN PHU		DONG AN PHU		TAY AN THOI		DONG BA DIEM		BEN NGHE								
			0		0		0		0		0		0		280				
			0		0		0		0		0		0		0				
			0		0		0		0		0		0		0				
			0		185		0		0		0		0		0				
			0		184		0		0		0		0		0				
			0		0		0		0		0		0		0				
			0		0		0		0		0		0		0				
			0		0		0		0		0		0		0				
			0		0		0		0		0		0		0				
			0		0		0		0		0		0		0				
			0		0		0		203		0		0		0				
			0		0		0		0		0		0		0				
			0		0		0		0		0		0		0				

	0	0	0	0	172	0	0
68	15	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	818	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
BEN THANH BINH AN BINH CHANH BINH CHIEU BINH HUNG BINH HUNG HOA							
291	0	0	0	0	0	0	
0	0	0	0	0	0	0	
0	0	0	0	0	0	0	
0	0	0	0	0	0	0	
0	292	0	0	0	0	0	
0	0	0	0	0	0	0	
0	0	0	0	0	0	0	
0	0	0	0	0	0	0	
0	0	0	0	0	0	0	
0	0	0	0	0	0	0	
0	0	0	0	0	0	0	
0	0	1	0	0	0	0	
0	0	341	0	974	0	0	
0	0	0	0	0	0	1870	
0	0	0	0	0	0	0	
0	0	0	0	0	0	1	
0	0	0	0	0	0	0	
0	0	0	0	0	0	0	
0	0	0	0	0	0	0	
0	0	0	0	0	0	0	
0	0	0	0	0	0	0	
0	0	0	0	0	0	0	
0	0	0	0	0	0	1	
0	0	0	614	0	0	0	
BINH HUNG HOA A BINH HUNG HOA B BINH KHANH BINH LOI BINH MY BINH THO							
	0	0	0	0	0	0	0
	0	0	0	0	0	0	0
	0	0	0	0	0	0	0
	0	0	0	0	0	0	0
	0	0	57	0	0	0	0
	0	0	0	0	0	0	0
	0	0	0	0	0	0	0
	0	0	0	0	0	0	0

0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	122	0	0
1138	653	0	0	0	0
0	0	0	0	0	0
0	0	371	0	0	0
0	0	0	0	71	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	206
BINH THUAN BINH TRI DONG BINH TRI DONG A BINH TRI DONG B BINH TRUNG DONG					
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	174
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
482	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	1512	557	372	0	0
0	0	0	0	0	0
0	0	1	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
BINH TRUNG TAY CAN THANH CAT LAI CAU KHO CAU ONG LANH CO GIANG DA KAO DA PHUOC					
0	0	0	312	206	344 279 0

0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
213	0	189	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	268
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	73	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
DONG HUNG THUAN	DONG THANH	HIEP BINH	CHANH	HIEP BINH	PHUOC	HIEP PHU	HIEP PHUOC
0	0		0		0	0	0
0	0		0		0	0	0
0	0		0		0	0	0
858	0		0		0	0	0
1	0		0		0	0	0
0	0		0		0	0	0
0	0		0		0	0	0
0	0		0		0	0	0
0	0		0		0	0	0
0	0		0		0	0	0
0	0		0		0	0	0
0	0		0		0	338	0
0	0		0		0	0	0
0	0		0		0	0	0
0	0		0		0	0	0
0	0		0		0	0	0
0	0		0		0	0	0
1	0		0		0	0	0
0	368		0		0	0	0

	0	0	0	0	0	0	215
	0	0	0	0	0	0	0
	0	0	0	0	0	0	0
	0	0	0	0	0	0	0
	0	0	758	458	0	0	0
HIEP TAN HIEP THANH HOA PHU HOA THANH HUNG LONG KHONG RO LE MINH XUAN							
0	0	0	0	0	1287	0	
0	0	0	0	0	947	0	
0	0	0	0	0	1348	0	
0	531	0	0	0	423	0	
0	0	0	0	0	444	0	
0	0	0	0	0	766	0	
0	0	0	0	0	1038	0	
0	0	0	0	0	1533	0	
0	0	0	0	0	3006	0	
0	0	0	0	0	1797	0	
0	0	0	0	0	5156	0	
0	0	0	0	0	268	0	
0	0	0	0	281	1619	313	
0	0	0	0	0	445	0	
0	0	0	0	0	1181	0	
0	0	0	0	0	204	0	
0	0	62	0	0	114	0	
0	1	0	0	0	692	0	
0	0	0	0	0	361	0	
0	0	0	0	0	713	0	
0	0	0	0	0	300	0	
0	0	0	0	0	1046	0	
473	0	0	383	0	445	0	
0	0	0	0	0	571	0	
LINH CHIEU LINH DONG LINH TAY LINH TRUNG LINH XUAN LONG BINH LONG HOA							
0	0	0	0	0	0	0	
0	0	0	0	0	0	0	
0	0	0	0	0	0	0	
0	0	0	0	0	0	0	
0	0	0	0	0	0	0	
0	0	0	0	0	0	0	
0	0	0	0	0	0	0	
0	0	0	0	0	0	0	
0	0	0	0	0	0	0	
0	0	0	0	0	0	0	
0	0	0	0	0	0	0	
0	0	0	0	0	229	0	

0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	76
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
237	260	248	696	743	0	0	0
LONG PHUOC	LONG THANH	MY LONG	THOI LONG	TRUONG LY	NHON NGUYEN	CU TRINH	
0		0	0	0	0		555
0		0	0	0	0		0
0		0	0	0	0		0
0		0	0	0	0		0
0		0	0	0	0		0
0		0	0	0	0		0
0		0	0	0	0		0
0		0	0	0	0		0
0		0	1	0	0		0
0		0	0	0	0		0
107	233	0	102	0	0		0
0	0	0	0	0	0		0
0	0	0	0	0	0		0
0	0	0	0	0	0		0
0	0	0	0	0	50		0
0	0	0	0	0	0		0
1	0	0	0	0	0		0
0	0	0	0	0	0		0
0	0	157	0	0	0		0
0	0	0	0	0	0		0
0	0	0	0	0	0		0
0	0	0	0	0	0		0
0	0	0	0	0	0		0
NGUYEN THAI	BINH NHI	BINH NHON	DUC NHUAN	DUC PHAM	NGU LAO	PHAM VAN	COI
	209	0	0	0	349		0
	0	0	0	0	0		0
	0	0	0	0	0		0
	0	0	0	0	0		0
	0	0	0	0	0		0

64

0	0	0	0	0	0	0
PHU THO HOA	PHU THUAN	PHU TRUNG	PHU XUAN	PHUOC BINH	PHUOC HIEP	PHUOC KIENG
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	2	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	426	0	0	0	0	1
0	0	0	0	0	0	0
0	0	0	0	203	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	1
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	52	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	497	0	0	356
0	0	0	0	0	0	0
0	0	0	0	0	0	0
762	0	483	0	0	0	0
0	0	0	0	0	0	0
PHUOC LOC	PHUOC LONG A	PHUOC LONG B	PHUOC THANH	PHUOC VINH	AN	Phường 1
0	0	0	0	0	0	0
0	0	0	0	0	0	232
0	0	0	0	0	0	175
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	131
0	0	0	0	0	0	81
0	0	0	0	0	0	144
0	0	0	0	0	0	202
0	0	0	0	0	0	0
0	0	0	0	0	0	240
0	267	492	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	75
0	0	0	0	0	0	0

0	0	0	77	99	0	
0	0	0	0	0	73	
0	0	0	0	0	0	
171	0	0	0	0	0	
0	0	0	0	0	86	
0	0	0	0	0	147	
0	0	0	0	0	0	
0	0	0	0	0	0	
Phường 10	Phường 11	Phường 12	Phường 13	Phường 14	Phường 15	Phường 16
0	0	0	0	0	0	0
183	161	439	306	514	381	0
247	275	190	202	301	205	187
0	0	0	0	0	0	0
0	0	0	0	0	0	0
127	300	288	95	210	0	0
136	0	0	470	198	197	215
422	194	261	180	193	0	0
469	430	424	467	312	0	0
0	0	0	0	0	0	0
478	133	243	143	319	572	510
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	552	673	935	218	491	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
441	409	852	139	421	160	345
0	0	0	0	0	0	0
0	0	0	0	0	0	0
284	220	0	237	0	156	0
1288	673	847	1415	673	1118	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
Phường 17	Phường 18	Phường 19	Phường 2	Phường 21	Phường 22	Phường 24
0	0	0	0	0	0	0
0	0	0	309	0	0	0
0	0	0	159	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	156	0	0	0
0	124	0	309	0	0	0
0	0	0	262	0	0	0
0	0	0	107	0	0	0

0	0	0	0	0	0	0	0
0	0	0	474	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
404	0	255	71	346	519	468	784
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
383	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
119	0	0	215	0	0	0	0
0	0	0	662	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
Phường 26	Phường 27	Phường 28	Phường 3	Phường 4	Phường 5	Phường 6	Phường 7
0	0	0	0	0	0	0	0
0	0	0	0	189	153	131	107
0	0	0	418	157	540	213	166
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	157	276	254	771	0
0	0	0	147	304	0	184	0
0	0	0	110	179	221	120	181
0	0	0	146	234	236	224	170
0	0	0	0	0	0	0	0
0	0	0	536	951	825	680	679
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
807	533	149	388	0	254	166	226
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	363	158	480	246	263
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	131	163	211	0	351
0	0	0	242	690	350	413	286
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
Phường 8	Phường 9	Phường An Lạc	Phường An Lạc A	Phường An Lợi Đông			
0	0	0	0	0		0	
114	244	0	0	0		0	

257	198	0	0	0
0	0	0	0	0
0	0	0	0	39
0	311	0	0	0
255	129	0	0	0
106	239	0	0	0
401	160	0	0	0
0	0	0	0	0
238	304	0	0	0
0	0	0	0	0
0	0	0	0	0
0	0	581	1106	0
0	0	0	0	0
0	0	0	0	0
0	0	0	0	0
248	213	0	0	0
0	0	0	0	0
0	0	0	0	0
121	347	0	0	0
369	419	0	0	0
0	0	0	0	0
0	0	0	0	0
Phường An Phú	Phường An Phú	Đông	Phường Bến Nghé	Phường Bến Thành
0		0	358	290
0		0	0	0
0		0	0	0
0		247	0	0
579		0	0	0
0		0	0	0
0		0	0	0
0		0	0	0
0		0	0	0
0		0	0	0
0		0	0	0
0		0	0	0
0		0	0	0
0		0	0	0
0		0	0	0
0		0	0	0
0		0	0	0
0		0	0	0
0		0	0	0
0		0	0	0
0		0	0	0

0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
Phường Bình An	Phường Bình Chiểu	Phường Bình Hưng Hoà	Phường Bình Hưng Hoà A
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
480	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	3262	3080
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	1295	0	0
Phường Bình Hưng Hoà B	Phường Bình Thọ	Phường Bình Thuận	Phường Bình Trị Đông
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	611	0
0	0	0	0
0	0	0	0
0	0	0	0

	2218	0	0	2030
	0	0	0	0
	0	0	0	0
	0	0	0	0
	0	0	0	0
	0	0	0	0
	0	0	0	0
	0	0	0	0
	0	0	0	0
	0	0	0	0
	0	174	0	0
Phường Bình Trị Đông A		Phường Bình Trị Đông B	Phường Bình Trưng Đông	
	0	0	0	
	0	0	0	
	0	0	0	
	0	0	0	
	0	0	379	
	0	0	0	
	0	0	0	
	0	0	0	
	0	0	0	
	0	0	0	
	0	0	0	
	0	0	0	
	0	0	0	
	0	0	0	
	1441	1161	0	
	0	0	0	
	0	0	0	
	0	0	0	
	0	0	0	
	0	0	0	
	0	0	0	
	0	0	0	
	0	0	0	
	0	0	0	
	0	0	0	
Phường Bình Trưng Tây		Phường Cát Lái	Phường Cầu Kho	Phường Cầu Ông Lãnh
	0	0	348	311
	0	0	0	0
	0	0	0	0
	0	0	0	0
	550	331	0	0
	0	0	0	0

[illegible]

Phường Hiệp Bình Phước	Phường Hiệp Phú	Phường Hiệp Tân	Phường Hiệp Thành
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	2208
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	294	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	798	0
760	0	0	0

Phường Hòa Thạnh	Phường Linh Chiểu	Phường Linh Đông	Phường Linh Tây
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0

0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
839	0	0	0
0	318	390	241
Phường Linh Trung	Phường Linh Xuân	Phường Long Bình	Phường Long Phước
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	205	92
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
929	673	0	0
Phường Long Thạnh Mỹ	Phường Long Trường	Phường Nguyễn Cư Trinh	
0	0	628	
0	0	0	
0	0	0	
0	0	0	
0	0	0	
0	0	0	
0	0	0	
0	0	0	
0	0	0	
0	0	0	

	0		0		0
	280		203		0
	0		0		0
	0		0		0
	0		0		0
	0		0		0
	0		0		0
	0		0		0
	0		0		0
	0		0		0
	0		0		0
	0		0		0
	0		0		0
	0		0		0
Phường Nguyễn Thái Bình	283	Phường Phạm Ngũ Lão	366	Phường Phú Hữu	0
	0		0		0
	0		0		0
	0		0		0
	0		0		0
	0		0		0
	0		0		0
	0		0		0
	0		0		0
	0		0		420
	0		0		0
	0		0	307	0
	0		0		0
	0		0		0
	0		0		0
	0		0		0
	0		0		0
	0		0		0
	0		0		0
	0		0		0
	0		0		0
	0		0		0
Phường Phú Thạnh	0	Phường Phú Thọ Hoà	0	Phường Phú Thuận	0
	0		0		0
	0		0		0

0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	605	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
1375	1572	0	1083
0	0	0	0
Phường Phước Bình	Phường Phước Long A	Phường Phước Long B	Phường Sơn Kỳ
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
187	321	991	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0

[illegible]

0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
792	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	1048	0	531
0	0	0	0
Phường Tân Thới Nhất	Phường Tân Thuận Đông	Phường Tân Thuận Tây	
0	0	0	
0	0	0	
0	0	0	
930	0	0	
0	0	0	
0	0	0	
0	0	0	
0	0	0	
0	0	0	
0	393	443	
0	0	0	
0	0	0	
0	0	0	
0	0	0	
0	0	0	
0	0	0	
0	0	0	
0	0	0	
0	0	0	
0	0	0	
0	0	0	
0	0	0	
0	0	0	
0	0	0	
0	0	0	
Phường Tăng Nhơn Phú A	Phường Tăng Nhơn Phú B	Phường Tây Thạnh	

	0		0		0
	0		0		0
	0		0		0
	0		0		0
	0		0		0
	0		0		0
	0		0		0
	0		0		0
	0		0		0
	0		0		0
	0		0		0
	0		0		0
	518		516		0
	0		0		0
	0		0		0
	0		0		0
	0		0		0
	0		0		0
	0		0		0
	0		0		0
	0		0		0
	0		0		0
	0		0		0
	0		0		0
	0		0		0
	0		0		1480
	0		0		0
Phường Thạnh Lộc	Phường Thạnh Mỹ Lợi	Phường Thạnh Xuân	Phường Thảo Điền		
0	0	0	0		
0	0	0	0		
0	0	0	0		
404	0	365	0		
0	439	0	349		
0	0	0	0		
0	0	0	0		
0	0	0	0		
0	0	0	0		
0	0	0	0		
0	0	0	0		
0	0	0	0		
0	0	0	0		
0	0	0	0		
0	0	0	0		
0	0	0	0		
0	0	0	0		
0	0	0	0		
0	0	0	0		
0	0	0	0		

0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
Phường Thới An	Phường Thủ Thiêm	Phường Trung Mỹ Tây	Phường Trường Thạnh
0	0	0	0
0	0	0	0
0	0	0	0
782	0	951	0
0	76	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	216
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
Phường Trường Thọ	QUY DUC SON KY TAM BINH TAM PHU TAM THON HIEP TAN AN		
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0

	0	0	0	0	0	0	0	0
	0	160	0	0	0	0	0	0
	0	0	0	0	0	0	0	0
	0	0	0	0	0	0	0	0
	0	0	0	0	0	39	5	0
	0	0	0	0	0	0	0	0
	0	0	0	0	0	0	0	0
	0	0	0	0	0	0	0	0
	0	0	0	0	0	0	0	0
	0	0	0	0	0	0	0	0
	0	0	0	0	0	0	0	0
	0	0	620	0	0	0	0	0
	498	0	0	277	180	0	0	0
TAN AN HOI	TAN CHANH HIEP	TAN DINH	TAN HIEP	TAN HUNG	TAN HUNG	THUAN	TAN KIEN	
0	0	367	0	0	0	0	0	
0	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	
0	511	0	0	0	0	231	0	
0	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	
0	0	0	0	0	685	0	549	
0	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	461	
0	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	
132	0	0	0	0	0	0	0	
0	1	0	0	0	0	0	0	
0	0	0	208	0	0	0	0	
0	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	
TAN NHUT	TAN PHONG	TAN PHU	TAN PHU TRUNG	TAN QUI TAY	TAN QUY	TAN SON	NHI	
0	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	

0	0	691	0	0	0
0	0	0	0	0	0
TAN THOI HOA	TAN THOI NHAT	TAN THOI NHI	TAN THONG HOI	TAN THUAN DONG	
0	0	0	0	0	
0	0	0	0	0	
0	0	0	0	0	
0	840	0	0	0	
0	0	0	0	0	
0	0	0	0	0	
0	0	0	0	0	
0	0	0	0	0	
0	0	0	0	0	
0	0	0	0	564	
0	0	0	0	0	
0	0	0	0	0	
0	0	0	0	0	
0	0	0	0	0	
0	0	1	0	0	
0	0	0	0	0	
0	0	0	172	0	
0	0	0	0	0	
0	0	207	0	0	
0	0	0	0	0	
0	0	0	0	0	
0	0	0	0	0	
374	0	0	0	0	
0	0	0	0	0	
TAN THUAN TAY	TAN TUC	TAN XUAN	TANG NHON PHU A	TANG NHON PHU B	TAY THANH
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
476	0	0	0	0	0
0	0	0	0	0	0
0	0	0	334	258	0
0	205	0	0	1	0
0	0	0	0	0	0
0	0	0	0	0	0

	0	0	0		0	0	0
	0	0	0		0	0	0
	0	0	0		0	0	0
	0	0	258		0	0	0
	0	0	0		0	0	0
	0	0	0		0	0	0
	0	0	0		0	0	0
	0	0	0		0	0	517
	0	0	0		0	0	0
THAI MY THANH LOC THANH MY LOI THANH XUAN THAO DIEN THI TRAN Thị Trấn Củ Chi	0	0	0	0	0	0	0
	0	0	0	0	0	0	0
	0	0	0	0	0	0	0
	0	227	0	201	0	0	0
	0	0	174	0	259	0	0
	0	0	0	0	1	0	0
	0	0	0	0	0	0	0
	0	0	0	0	0	0	0
	0	0	0	0	0	0	0
	0	0	0	0	0	0	0
	0	0	0	0	0	0	0
	0	0	0	0	0	0	0
	0	0	0	0	0	0	0
	0	0	0	0	0	0	0
	0	0	0	0	0	0	0
	0	0	0	0	0	1	0
35	0	0	0	0	0	146	566
	0	0	0	0	0	0	0
	0	0	0	0	0	357	0
	0	0	0	0	0	353	0
	0	0	0	0	0	0	0
	0	0	0	0	0	0	0
	0	0	0	0	0	0	0
	0	0	0	0	0	0	0
Thị trấn Hóc Môn Thị Trấn Nhà Bè Thị trấn Tân Túc THOI AN THOI TAM THON	0	0	0	0	0	0	0
	0	0	0	0	0	0	0
	0	0	0	0	0	0	0
	0	0	0	0	293	0	0
	0	0	0	0	0	0	0
	0	0	0	0	0	0	0
	0	0	0	0	0	0	0
	0	0	0	0	0	0	0

0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
Xã Bình Khánh	Xã Bình Lợi	Xã Bình Mỹ	Xã Cần Thạnh	Xã Đa Phước	Xã Đông Thạnh
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	85	0	0	124	0
0	0	0	0	0	0
0	0	0	0	0	0
217	0	0	185	0	0
0	0	281	0	0	0
0	0	0	0	0	0
0	0	0	0	0	1243
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
Xã Hiệp Phước	Xã Hòa Phú	Xã Hưng Long	Xã Lê Minh Xuân	Xã Long Hòa	Xã Long Thới
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0

0	0	362	357	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	237	0
0	241	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
249	0	0	0	0	264
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
Xã Lý Nhơn Xã Nhị Bình Xã Nhơn Đức Xã Nhuận Đức Xã Phạm Văn Cội					
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
45	0	0	0	0	0
0	0	0	73	51	0
0	0	0	0	0	0
0	159	0	0	0	0
0	0	296	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
Xã Phạm Văn Hai Xã Phong Phú Xã Phú Hòa Đông Xã Phú Mỹ Hưng Xã Phú Xuân					
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0

Xã Phước Hiệp Xã Phước Kiển Xã Phước Lộc Xã Phước Thạnh Xã Phước Vĩnh An

	0		0		0		0		0
Xã Quy Đức		Xã Tam Thôn Hiệp		Xã Tân An Hội		Xã Tân Hiệp		Xã Tân Kiên	Xã Tân Nhựt
0		0		0		0		0	0
0		0		0		0		0	0
0		0		0		0		0	0
0		0		0		0		0	0
0		0		0		0		0	0
0		0		0		0		0	0
0		0		0		0		0	0
0		0		0		0		0	0
0		0		0		0		0	0
0		0		0		0		0	0
77		0		0		0		532	203
0		0		0		0		0	0
0		0		0		0		0	0
0		113		0		0		0	0
0		0		542		0		0	0
0		0		0		0		0	0
0		0		0		558		0	0
0		0		0		0		0	0
0		0		0		0		0	0
0		0		0		0		0	0
0		0		0		0		0	0
0		0		0		0		0	0
Xã Tân Phú Trung		Xã Tân Quý Tây		Xã Tân Thạnh Đông		Xã Tân Thạnh Tây			
0		0		0		0			
0		0		0		0			
0		0		0		0			
0		0		0		0			
0		0		0		0			
0		0		0		0			
0		0		0		0			
0		0		0		0			
0		0		0		0			
0		0		0		0			
0		0		0		0			
0		318		0		0			
0		0		0		0			
0		0		0		0			
0		0		0		0			

Xã Tân Thới Nhì Xã Tân Thông Hội Xã Tân Xuân Xã Thái Mỹ Xã Thạnh An

0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	358	0	142
0	0	0	0
1435	0	549	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
Xã Trung Lập Thượng	Xã Vĩnh Lộc A	Xã Vĩnh Lộc B	Xã Xuân Thới Đông
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	2865	2609	0
0	0	0	0
0	0	0	0
0	0	0	0
161	0	0	0
0	0	0	0
0	0	0	550
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
Xã Xuân Thới Sơn	Xã Xuân Thới Thượng	XUÂN THOI DÔNG	XUÂN THOI SƠN
0	0	0	0
0	0	0	0

0
0
0
0

```
s3_cleaned_incidence_dat <- s2_cleaned_incidence_dat |>
mutate(
  commune = stri_trans_general(commune, id = "Latin-ASCII") %>% tolower(),
  commune = str_remove(commune, "phuong|xa|thi tran") %>% trimws(),
  commune = trimws(commune, which = "left", whitespace = "0"),
  commune = case_when(
    commune == "khong ro" ~ NA,
    commune == "" ~ NA,
    .default = commune
  )
)
```

```
s4_cleaned_incidence_dat <- s3_cleaned_incidence_dat |>
select(sex, age, date, district, commune, new_commune, in_out_patient) |>
rename(
  date_admitted = date,
  status = in_out_patient
)
```

Export to CSV

```
s4_cleaned_incidence_dat |>
  arrange(date_admitted) |>
  write_csv("out/incidence_clean_2000_2020.csv")
```

Data viz

Raw time series

```
raw_ts_plot_dat <- s4_cleaned_incidence_dat %>%
  group_by(across(any_of(c("district", "date_admitted")))) %>%
  tally() %>%
```

```

complete(date_admitted, fill = list(n = 0)) %>%
mutate(
  year = year(date_admitted),
  isoweek = isoweek(date_admitted),
  # isoweek = ifelse(isoweek == 53, 52, isoweek)
) %>%
group_by(year, isoweek) %>%
summarise(date = min(date_admitted), n = sum(n))

raw_ts_plot_dat %>%
  ggplot(aes(x = date, y = n)) +
  geom_step() +
  # facet_wrap(~district, scales = "free_y", ncol = 4) +
  scale_y_continuous("Weekly reported dengue incidence") +
  scale_x_date(
    "",
    date_breaks = "4 years",
    date_labels = "%Y"
  )

raw_ts_plot_dat %>%
  ggplot(aes(x = date, y = n)) +
  # covid lockdown period
  annotate(
    "rect",
    xmin = as.Date("2020/01/01"),
    xmax = as.Date("2022/01/01"),
    ymin = 0,
    ymax = max(raw_ts_plot_dat$n),
    fill = "red",
    alpha = 0.5
  ) +
  geom_vline(
    xintercept = as.Date("2020/01/01"),
    linetype = "dashed",
    alpha = 0.5
  ) +
  annotate(
    "label",
    x = as.Date("2020/01/01"),
    y = 3400,
    label = "COVID-19\nlockdowns",

```

```

    color = "#ff4946"
  ) +
  geom_step() +
  # circ 54
  annotate(
    "segment",
    x = as.Date("2018/01/01"),
    y = 0,
    yend = 2800,
    linetype = 2,
    color = "blue"
  ) +
  annotate(
    "label",
    x = as.Date("2018/01/01"),
    y = 2800,
    label = "Circular\n54/2017/TT-BYT\nin effect",
    color = "blue"
  ) +
  scale_y_continuous("") +
  scale_x_date(
    "",
    date_breaks = "2 years",
    date_minor_breaks = "1 year",
    date_labels = "%Y"
  ) +
  ggtitle("Weekly reported dengue incidence in HCMC")

ggsave("plots/raw_ts_plot.jpg", width = 10)

```

Incidence by case status

```

case_status_plot_dat <- s4_cleaned_incidence_dat %>%
  group_by(date_admitted, status) %>%
  tally() %>%
  ungroup() %>%
  complete(date_admitted, status, fill = list(n = 0)) %>%
  mutate(
    year = year(date_admitted),
    isoweek = isoweek(date_admitted),

```

```

    # isoweek = ifelse(isoweek == 53, 52, isoweek)
  ) %>%
  group_by(year, isoweek, status) %>%
  summarise(date = min(date_admitted), n = sum(n))

```

``summarise()`` has grouped output by 'year', 'isoweek'. You can override using the ``.groups`` argument.

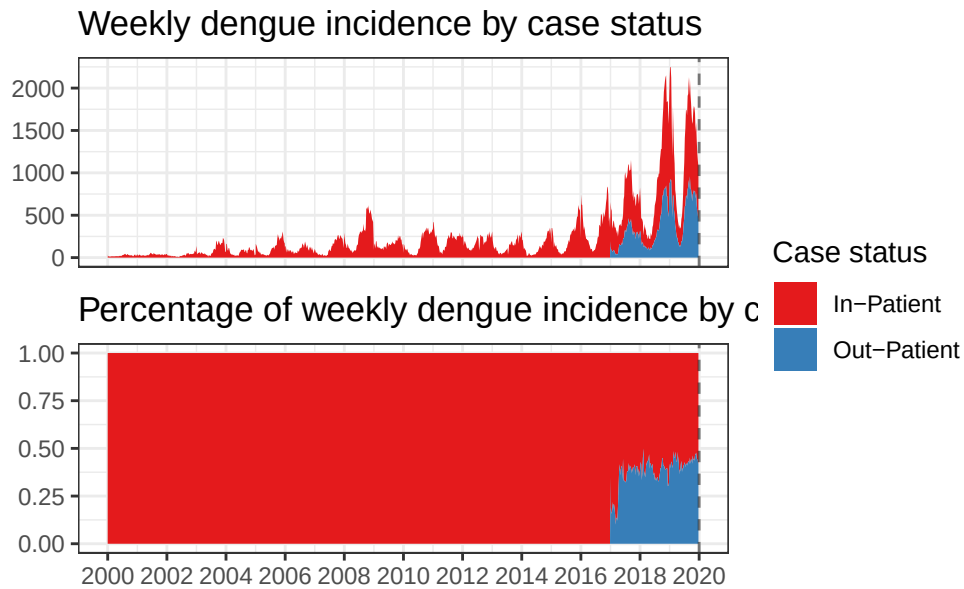
```

p1 <- case_status_plot_dat %>%
  ggplot(aes(x = date, y = n, fill = status)) +
  geom_area() +
  ggtitle("Weekly dengue incidence by case status")

p2 <- case_status_plot_dat %>%
  group_by(date) %>%
  mutate(n_pct = n / sum(n)) %>%
  ggplot(aes(x = date, y = n_pct, fill = status)) +
  geom_area() +
  ggtitle("Percentage of weekly dengue incidence by case status")

(p1 / p2) +
  plot_layout(guides = "collect", axes = "collect") &
  geom_vline(
    xintercept = as.Date("2020/01/01"),
    linetype = "dashed",
    alpha = 0.5
  ) &
  scale_fill_brewer(
    "Case status",
    labels = str_to_title,
    palette = "Set1"
  ) &
  scale_x_date(
    "",
    date_breaks = "2 years",
    date_minor_breaks = "1 year",
    date_labels = "%Y"
  ) &
  scale_y_continuous("")

```



```
ggsave("plots/case_status_plot.jpg", width = 10)
```

Saving 10 x 3.5 in image

Data availability map

```
s4_cleaned_incidence_dat %>%
  mutate(week = floor_date(date_admitted, "week")) %>%
  group_by(week, status, district) %>%
  tally() %>%
  ungroup() %>%
  complete(week, status, district) %>%
  filter(status %in% c("in-patient", "out-patient")) %>%
  group_by(week, status, district) %>%
  mutate(total_n = sum(n, na.rm = TRUE)) %>%
  ungroup() %>%
  mutate(hospital = fct_lump_prop(district, prop = 0.01, w = total_n)) %>%
  ggplot(aes(x = week, y = fct_reorder(district, total_n), fill = n)) +
  geom_raster() +
  facet_wrap(~status, ncol = 1) +
```

```

theme(legend.position = "none") +
scale_x_date(
  date_breaks = "1 year",
  date_labels = "%Y",
  minor_breaks = NULL
) +
scale_fill_viridis_c(na.value = "transparent")

```

